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# High-throughput alloy and process design for metal additive manufacturing

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Many engineering alloys originally designed for conventional manufacturing lack considerations for additive manufacturing (AM), presenting opportunities for novel alloy designs. Evaluating alloy printability requires extensive analysis of chemical composition and processing conditions. The complexity of experimental exploration drives the need for high-throughput computational frameworks. This study introduces a framework that integrates material properties, processing parameters, and melt pool profiles from three thermal models to assess process-induced defects, such as lack-of-fusion, balling, and keyholing. A deep learning surrogate model accelerates the printability assessment by 1000 times without losing accuracy. We validate the framework with printability maps for the equiatomic CoCrFeMnNi system and apply it to explore printable alloys in the Co-Cr-Fe-Mn-Ni high-entropy alloy space. Ensemble probabilistic printability maps further provide insights into defect likelihood and uncertainty, enhancing alloy design for AM by efficiently navigating vast design spaces.

Additive manufacturing (AM) is rapidly emerging as a pivotal technology for industrial part production due to its ability to create intricate shapes through computer-assisted layer-by-layer part construction[1]. In metal AM, a broad, ever-expanding technology landscape has emerged with varying degrees of adoption in industry[2]. While various AM technologies exist, liquid-mediated metal AM, notably Laser Powder Bed Fusion (L-PBF), has garnered significant interest in industry and academia due to its potential to produce high-quality parts. L-PBF's quality relies on understanding the interplay between process parameters, material properties, and solidification conditions.

Traditional alloy design primarily targets specific manufacturing routes[3–5] and often overlooks the unique demands of additive manufacturing (AM). In AM, designing alloys requires optimizing both chemistry and processing conditions simultaneously to minimize defects and enhance reliability. Conventional approaches first focus on achieving target properties and then on developing scalable processing methods. However, the tight coupling between process and chemistry in AM requires simultaneous optimization of alloy composition and process parameters. Key factors include process parameters (e.g., laser power, scan speed, hatch spacing), material properties (e.g., thermal conductivity, heat capacity), and performance metrics (e.g., strength, ductility, corrosion resistance).

When considering the manufacturability of alloys under L-PBF AM, several different defect modes should be accounted for[6]:

- **Lack-of-fusion** arises either from the lack of powder melting (due to insufficient energy input) in the previous layer or from inadequate overlap between the adjacent tracks or layers resulting from large hatch spacing or layer thickness[7,8].
- **Balling** results from insufficient energy input, which creates a lack of molten liquid with high surface tension and reduces the liquid's wettability on the substrate[7,9].
- **Keyholing** occurs when excessive energy input causes porosities due to the high elemental evaporation[7,10] and generates deep liquid penetration as evaporation-induced recoil pressure acts on the melt pool surface[11,12].
- **Cracking** originates from a combination of high thermal gradient and residual stresses[7,8], while hot tearing (cracking), a material-dependent defect, results from obstructed thermal contraction (shrinkage)[13].

Given the tight coupling between thermal histories, thermo-physical properties, and defect-inducing phenomena, varying process parameters affect the likelihood of specific printing defects. Thus, printability maps[6] serve as valuable tools to identify defect-prone process parameter combinations, delineate at-risk regions, and guide the selection of optimal conditions to enhance AM design (see Fig. 1). These maps incorporate the maximum hatch spacing (Equation (13) in Table 7), which the melt pool geometry determines to ensure sufficient overlap between tracks in multi-

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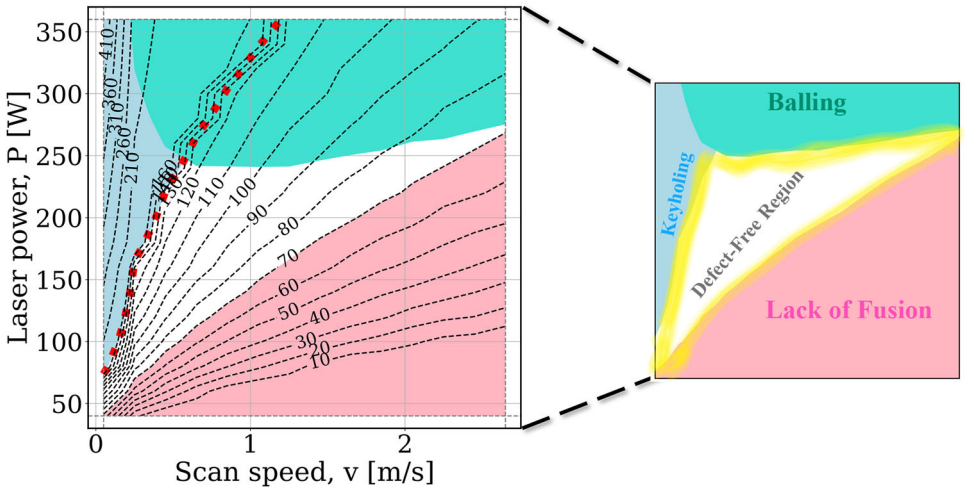



**Fig. 1 | An Example Printability Map.** This figure presents a general printability map for L-PBF metal AM processing conditions, highlighting macroscopic defects: lack-of-fusion (salmon), keyholing (blue), and balling (teal). The white area represents the predicted defect-free P-V space. Defect boundaries are based on specific criteria: the dashed black line indicates the maximum hatch spacing (Equation (13)) for adequate track overlap, and the red dashed line marks the keyholing boundary. Salmon: Lack-of-fusion predicted region. Blue: Keyholing predicted region. Teal: Balling predicted region. LOF lack-of-fusion, KH keyholing; Ball, balling.

scan prints. In this work, however, we focus solely on single-track scenarios, even though prior studies have shown that single-track printability maps can effectively identify optimal processing windows for fabricating defect-free 3D coupons[14] and more complex testing specimens[15].

To date, experiments, simulations, and machine learning (ML) approaches have successfully combined to assess the printability map for a given alloy chemistry[14–18]. However, transforming printability assessment into a tool for exploring vast alloy spaces remains a significant challenge. The high cost of manufacturing AM powder feedstock severely limits high-throughput (**HTP**) determination of printability maps for large numbers of alloys, especially when relying on experimental approaches. In our previous work, we showed that creating a printability map for a single composition using experimental calibration could take up to 10 days[19]. To accelerate printability map construction for HTP design, a fully computational framework must predict both *intrinsic* (due to thermo-physical properties) and *extrinsic* (induced by process parameters and thermal boundary conditions) susceptibility to defect formation in alloys of arbitrary compositions. Such a framework can accelerate the AM design process by identifying intrinsically printable alloys and their corresponding optimal processing windows.

Identifying appropriate criteria to differentiate defect regions in the composition and process spaces poses a key challenge. To address this challenge, in this work, we deploy alloy design criteria for intrinsic susceptibility to balling and hot cracking to guide design in the *composition space*. After identifying candidate alloy chemistries, we apply criteria based on materials properties, processing parameters, and thermal history proposed in the literature[7,12,14,15,20,21] to find the defect (lack-of-fusion, balling, keyholing) and defect-free regions in the *process space*.

In this work, we demonstrate the co-design of alloy chemistry and AM processing parameters using an exclusively computational approach. Our framework combines fast-acting models for the thermal history of a melt pool under arbitrary processing conditions, CALPHAD-based models for the phase stability and thermo-physical properties of alloys, and criteria for the onset of both intrinsic (e.g., hot-cracking susceptibility) and extrinsic defects (keyholing, lack-of-fusion). We designed the framework for HTP applications. Given its considerable interest by the scientific community, we demonstrate the framework over the 'Cantor' high-entropy alloy space. To establish the framework's validity, we validate our predictive printability assessment using experimental data from the stoichiometric Cantor alloy CoCrFeMnNi. We introduce a metric, *printability index*, to quantify an alloy's printability across diverse processing conditions. We then employ the framework to identify promising regions in this FCC HEA space with minimal susceptibility to fabrication defects. Figure 2 summarizes the key challenges, research gaps, and the proposed framework.

Developing printability maps for alloys requires criteria linking melt pool characteristics, powder bed attributes, and thermal histories to delineate defect-prone regions and identify the printable window within the process space. A fast and precise thermal model, such as the Eagar-Tsai (E-T) model[22], is essential for analyzing defect criteria in high-throughput workflows. Studies[6,11,14,15,20,23] have validated the E-T model for AM applications. Tapia et al.[23] demonstrated its ability to predict melt pool uncertainties in 430F stainless steel using polynomial chaos expansion. Johnson et al.[6] found E-T predictions for melt pool geometries comparable to high-fidelity COMSOL models, although discrepancies arose in process space regions corresponding to keyholing, as the E-T model only accounts for conduction heat transfer. Honarmandi et al.[11] addressed these discrepancies by benchmarking the E-T model against extensive datasets and incorporating keyhole corrections based on the Gladush and Smurov theory[24] and found much better agreement with experiments across broad ranges of processing parameters.

Recent studies have leveraged the E-T model to produce printability maps for various material systems such as Ni-5wt.%Nb alloy[6], Co-Cr-Fe-Mn-Ni high entropy alloy (HEA)[6,14], AF96 low alloy martensitic steel[14,15], and Ni-Ti alloys[20], printed via L-PBF. For instance, Islam et al.[25] employed a simplified analytical thermal model to construct a printability map for a Ni-based superalloy. However, in these works, researchers performed many L-PBF experiments across the process parameter space to calibrate the model and identify the criteria for different defect modes. The reliance on experiments makes these approaches unsuitable for HTP analysis. To reduce experimental dependency, researchers have introduced universal scaling laws based on dimensionless numbers.

One notable example is the dimensionless number $\Pi_1 = \frac{C_p P}{k v^2 h}$, which Rankouhi et al.[26] proposed and Islam et al.[25] later applied. Their work used this scaling law-and a corresponding law for relative density-to differentiate printable (defect-free) regions from defect regions. However, because they adopted material properties from the literature for a specific nickel-based superalloy, the framework is inherently limited to materials with available experimental data. Additionally, by relying on porosity-induced density changes, this approach only differentiates lack-of-fusion and keyholing regions, leaving phenomena such as balling unaddressed.

Zhu et al.[7] applied the scaling law introduced by Rubenchik et al.[27] to predict melt pool dimensions during the L-PBF AM processes as functions of two dimensionless variables—i.e., $p = \frac{D}{a v}$ and $B = \frac{\Delta H}{2^{3/4}\pi h_s}$, where $D$, $a$, and $\Delta H/h_s$ are the thermal diffusivity, the laser beam radius, and the normalized enthalpy, respectively. This work is fully computational and provides the closest comparison to the framework that we propose in this study. However, while Zhu et al.'s approach employs a single-scaled thermal model and one set of defect criteria for a specific material (a Ni-Ti shape memory alloy), our method integrates three different thermal models and uses twelve different combinations of defect criteria to generate printability maps for an

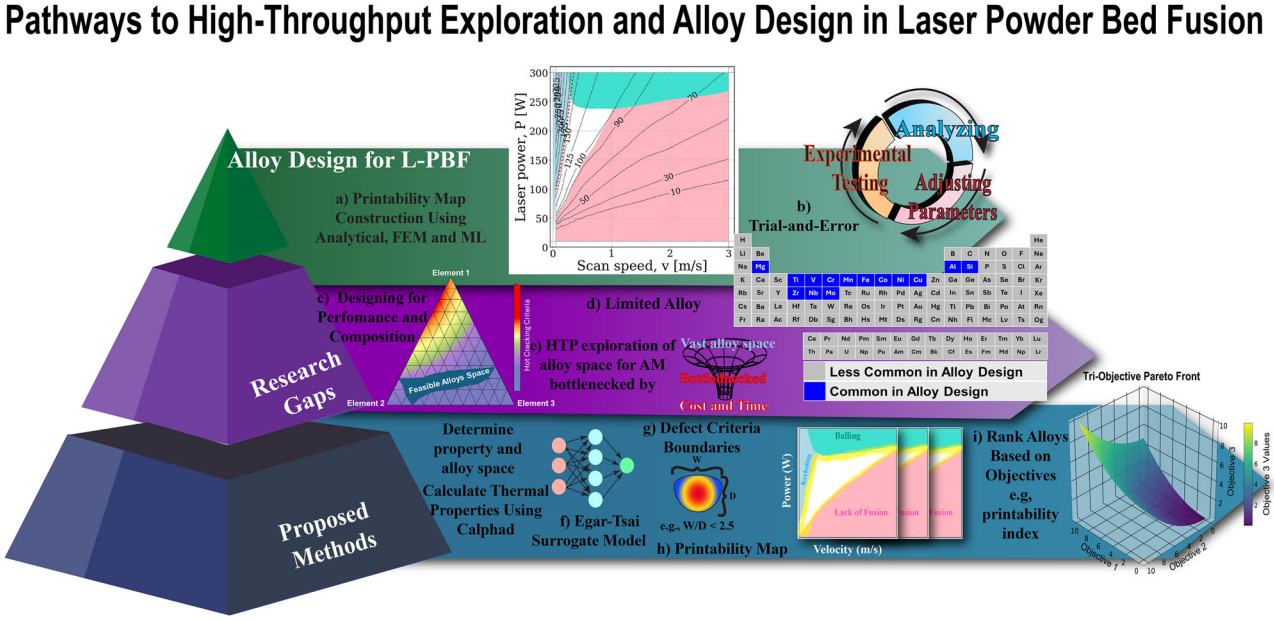



**Fig. 2 | Overview of the Challenges and Framework for Alloy Design in L-PBF.** Overview of challenges and research gaps in alloy design for L-PBF, alongside the proposed method. **Current Methods: a** Printability maps constructed using analytical, FEM, and ML approaches are computationally expensive due to the need for experimental validation. **b** Trial-and-error and DOE methods explore the alloy and composition space in L-PBF, resulting in significant costs in both time and expenses. **Research Gaps: c** Designing alloys that optimize both performance and composition. **d** Overcoming limitations of traditional methods and avoiding narrow alloy space exploration. **e** Mitigating bottlenecks in HTP exploration caused by limited tools and resources for alloy design in L-PBF. **Proposed Framework: f** Calculate material properties using CALPHAD methods and accelerate melt pool predictions through a surrogate model for the analytical thermal model. **g** Determine defect boundaries based on various criteria. **h** Use these boundaries to create printability maps. **i** Assess the printability of selected alloys and rank them based on property and printability criteria to achieve specific performance goals. FEM: finite element method. DOE: design of experiments. P [W]: laser power in watts. v [mm/s]: scan speed. W: melt pool width. D: melt pool depth.

entire alloy system (Cantor alloys), offering a more comprehensive and versatile analysis.

Another approach to accelerate printability map production for HTP AM design is applying ML models. Scime and Beuth[28] applied feature extraction methods and unsupervised ML approaches to detect different melt pool morphologies corresponding to different types of print defects. They indicated their prevalence across the process space for the Inconel 718 alloy. Du et al.[29] implemented a physics-informed ML model trained against binary data, i.e., "balling" or "no balling" occurrence, for six different alloys to predict balling susceptibility indices and maps in terms of six relevant mechanistic variables. Notably, these models target specific material systems with few training data points. Furthermore, Akbari et al.[30] used ML regression and classification via the MeltpoolNet package to predict melt pool geometry and defect modes; however, their results for printability maps show insufficient predictability compared to those derived from experiments, likely due to the sparsity of the training data and the challenges NNs face with sparse datasets. In our work, we train a neural network (NN) model with many relevant material properties (i.e., thermal conductivity, specific heat, etc.) using an extensive and validated computational database for different material systems to connect the chemistry and processing conditions to melt pool geometry during the AM processes in a material-agnostic manner. These relevant material properties help estimate the melt pool dimensions of arbitrary alloys. We then use these melt pool dimensions are then used to construct printability maps based on the defect criteria.

To address the issues above, in this work, we have developed a computational package to accelerate the construction of printability maps for the HTP design of defect-free AM parts with desired properties and performance. This package includes:

- **Material Properties:** Derived from CALPHAD-based methods and the rule-of-mixtures (see **Calculation of Thermophysical Properties**).
- **Thermal Models:** Incorporates the analytical E-T model, its dimensionless form, and a neural network surrogate for the E-T model.
- **Defect Criteria:** Implements two criteria for lack-of-fusion, two for balling, and three for keyholing to delineate defect and defect-free regions in the printability maps.
- **Map Construction:** Generates printability maps (twelve per composition for each defect combination).

To demonstrate our approach, we selected the stoichiometric CoCrFeMnNi (the equiatomic Cantor alloy) as our case study, given its status as one of the most well-investigated compositionally complex systems in AM. The Cantor alloy exhibits exceptional fracture toughness and ductility[31] and has attracted significant interest from the scientific community. This makes it an ideal representative for exploring the broader applications and optimizations possible within the High Entropy Alloy (HEA) category. More generally, HEAs in this chemical space-including those modified by the addition of elements such as Cu and Al-have garnered significant attention for their outstanding mechanical properties[32–48], high-temperature performance[49], and excellent corrosion and oxidation resistance[33,50].

Over the past decade, various trial-and-error strategies have explored processing parameters-such as laser power, scan speed, hatch spacing, laser beam diameter, and powder layer thickness-in printing HEAs[33,35,39,48,51–59]. Typically, the Archimedes density serves as a design constraint, while the volume energy density (VED) acts as a design indicator. The VED calculation follows: $VED = \frac{P}{v \cdot h \cdot t}$, where $P$ represents the laser power, $v$ the scan speed, $t$ the powder layer thickness, and $h$ the hatch spacing[35,36,41,42,44,53,54,60–62]. Optimal process conditions often align with specific VED ranges, although these ranges depend on the material.

Efforts to enhance the microstructure and mechanical properties of additively manufactured HEAs incorporate the addition of interstitial atoms (e.g., carbon and nitrogen) to improve mechanical properties[45,46,57,63,64] and employ nano-particle-based inoculants to refine the solidifying microstructure, thereby improving printability and mechanical behavior [61]. Rapid solidification during L-PBF induces thermal stresses that render HEAs susceptible to hot cracking[32,35,55,60,65–67]. To mitigate these issues, researchers

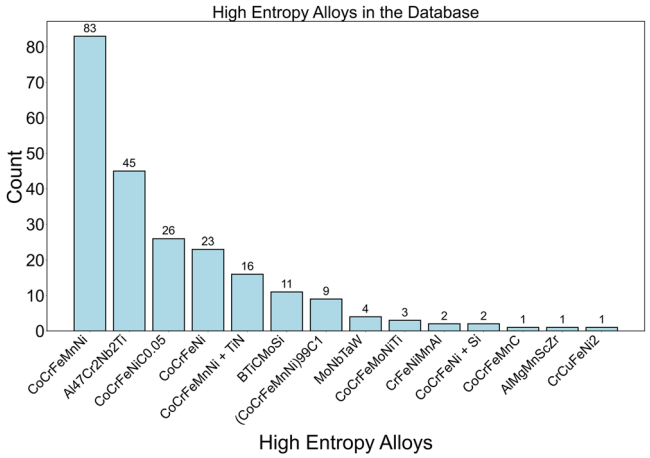



**Fig. 3 | Count of Literature Studies on High Entropy Alloys.** Studies on various HEAs in the literature show that HEAs containing Co, Cr, Fe, Mn, and Ni are among the most widely researched alloys. Therefore, we can narrow our composition space to include only HEAs with the specified elements.

have explored strategies such as substrate heating[51,55] and modifying scan strategies[62,63,67,68]. Additionally, various studies have applied post-processing techniques-such as annealing[42,47,69] and hot isostatic pressing[53]. Despite these efforts, challenges in fully eliminating hot cracking and achieving defect-free prints remain.

Beyond trial-and-error approaches, different groups have focused on developing more systematic approaches for the identification of optimal AM process conditions in the HEA space. For example, Chen et al.[44] proposed processing maps, and Johnson et al.[6] employed finite element modeling (FEM) in conjunction with a simpler Eagar-Tsai (E-T) thermal model to predict the printability of CoCrFeMnNi while quantifying uncertainties in defect criteria. Johnson et al.[6], however, relied on a single set of criteria for the onset of defect formation and used experimental measurements of thermophysical properties to parameterize the R-T model. While promising, the reliance of these frameworks on exhaustive experimental datasets challenges their transferability and renders their applicability in an HTP setting impractical.

Having selected the CoCrFeMnNi Cantor alloy as our case study, we utilized an internal database containing process conditions and printing outcomes, including defect types, to compare different sets of commonly used criteria for defect onset (lack of fusion, keyhole, balling), denoted as [LOFM-KHN-BallO], where [M, N, O] are indices for the different criteria for specific defect modes in Table 7. Each set of criteria can be considered to represent a distinct *model* for predicting defect formation. We assessed the accuracy of these models—constructed from melt pool dimensions predicted by different approximations of the E-T thermal model and other process-related metrics—by comparing their predictions against experimentally determined printing outcomes across broad regions of the L-PBF processing space using an in-house developed database. The accuracy metrics translated into model probabilities, which were in turn used to construct probabilistic printability maps. From these maps, we extracted a *printability index* representing the fraction of the processing space that leads to defect-free prints.

After identifying the optimal criteria set, we expanded our analysis to the broader Co-Cr-Fe-Mn-Ni compositional space by screening 3876 alloys for a stable FCC phase at 1300 °C, minimal susceptibility to hot cracking, and minimal susceptibility to balling defects. This screening process identified five alloys within the Co-Cr-Fe-Mn-Ni system for further analysis. For each of these alloys, we generated printability maps and calculated a *printability index*. By evaluating each alloy's susceptibility to balling, solidification range, susceptibility to hot-cracking, and printability index, we ranked the alloys to guide the co-design of alloy compositions and processing parameters optimized for defect-free AM parts.

In the following sections, we detail (i) the HTP printability map prediction using multiple criteria sets; (ii) how we combined these maps with experimental data to obtain probabilistic printability maps that reflect the accuracy of different criteria sets; (iii) how we extracted the printability index indicating the fraction of defect-free processing space; and (iv) how we integrated additional chemistry-dependent metrics to identify optimal alloys (retaining FCC structure at application-relevant temperatures) based on a four-factor performance metric, including the printability index derived from the ensemble probability printability map.

## Results

### Printability of an arbitrary Co-Cr-Fe-Mn-Ni high entropy alloy system

As mentioned above, we chose the equiatomic CoCrFeMnNi alloy to establish a ground truth to benchmark our study, as extensive studies in L-PBF have positioned it among the most thoroughly studied HEAs. Figure 3 shows this data, and the alloy exhibits a single-phase face-centered cubic (FCC) phase structure that provides good mechanical properties[60].

We constructed an in-house database of L-PBF alloys by comprehensively mining data in the literature and incorporating in-house experimental data. Each entry in the database captures the chemistry-processing space with over 500 features related to different aspects of metal L-PBF AM that affect the build. The features range from processing conditions, material properties, and powder properties to the printing outcome, post-performance of the printed part, and more. For this study, we filtered the database to focus on HEAs that meet specific criteria. We included only HEAs manufactured via L-PBF, with data detailing process parameters such as power and scan speed, as well as the final build state, including observed defects like lack-of-fusion, keyholing, and balling. Additionally, we considered only data points with experimental results that the literature has reported. The resulting subset data showcased in Fig. 3 reflects the availability and quality of the data rather than a selective focus on a specific system. The Cantor alloy system and its derivatives emerged prominently within this subset due to their prevalence in L-PBF literature and their well-characterized behavior under varying process conditions.

Figure 4 shows the combination of processing parameters for the CoCrFeMnNi system captured within the database.

Based on the data, we chose the range of processing parameters where the power was set from 40 W to 360 W, and the velocity was set from 0.05 m/s to 2.5 m/s. Furthermore, due to the limited data available, we considered single tracks and full builds to build a comprehensive data set that includes all the macroscopic defects mentioned (i.e., lack-of-fusion, keyholing, and balling).

For the equiatomic CoCrFeMnNi alloy, we constructed printability maps for each thermal model using 12 different combinations of defect criteria for lack-of-fusion (LOF), keyholing (KH), and balling (Ball). To simplify the identification of each criteria, we labeled each combination with a shorthand notation such as 'LOF2-KH2-Ball1', which specifies the corresponding defect criteria from Table 7. The numerical suffix in the shorthand indicates the specific equation that corresponds to each defect type (e.g., LOF2 refers to the second criterion listed in Table 7 for lack-of-fusion, as Equation (12) defines it). **Summary of Processing-Informed Criteria** discusses these defect criteria in more detail in Table 7. As detailed in **Methods**, we evaluated the accuracy (overall correctness), precision (reliability), and recall (sensitivity) of each criterion set by comparing the resulting printability map with literature-reported printing outcomes. To provide a complete analysis of the performance of the individual printability maps, we also present each map's precision and recall values in the Supplementary Material.

The thermal E-T model uses a 17-by-14 design of experiments for power and velocity. However, the dimensionless E-T thermal model uses a reduced processing space to keep the values of $p$ and $B$ (scaling parameters used in the dimensionless E-T model) within the range of 0.1-5 and 1-20, respectively, in these calculations. Based on the extracted experimental data, we set the hatch spacing, beam diameter, and powder layer thickness at

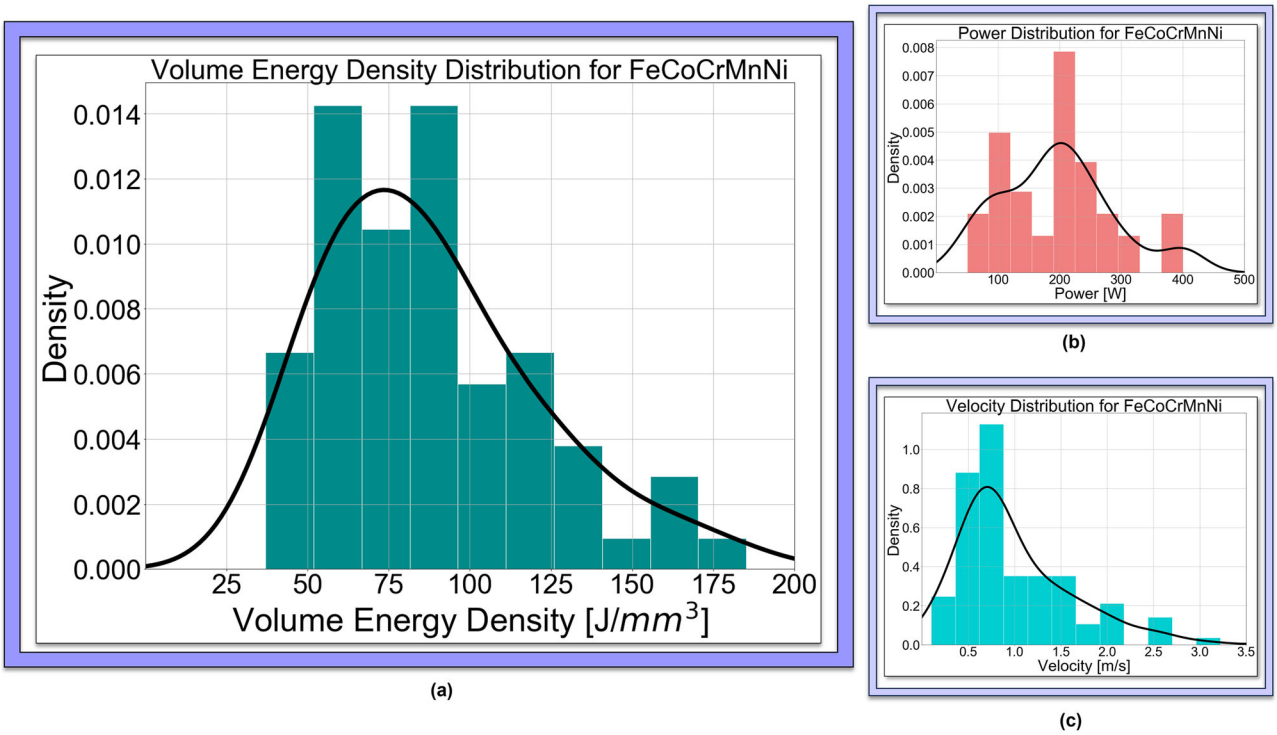



**Fig. 4 | Parameter spaces explored in literature for cantor alloy.** The histogram and kernel density estimate (KDE) plots for **a** volume energy density (VED), **b** power, and **c** velocity show the distribution of values used to print for the equiatomic CoCrFeMnNi alloy, as documented in the literature. The density represents the probability density function of the parameter on the x-axis in the plot.

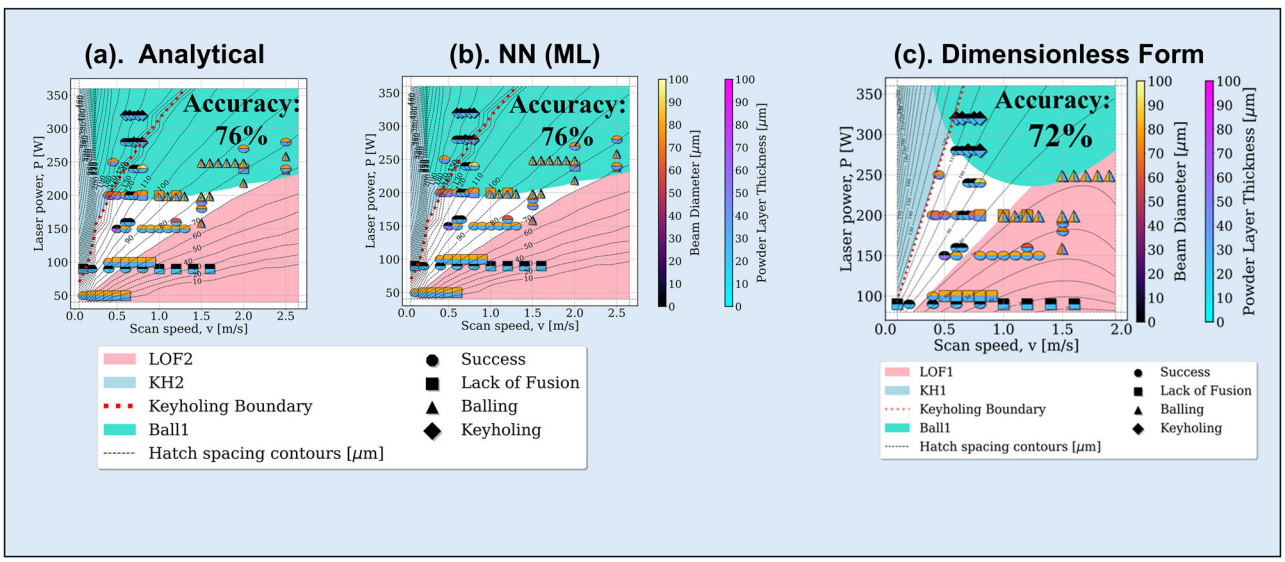



**Fig. 5 | Printability Maps with Highest Accuracy for the Cantor Alloy System Using Each Thermal Model. a** Using the **LOF2, KH2, and Ball1** criteria set and the analytical E-T thermal model, the predicted printability map exhibited an accuracy of 76%. The corresponding printable index is 17.1%. **b** Using the NN (ML) E-T model to calculate the melt pool geometry, the accuracy was again 76% while the printability index was calculated to be 18.5%. **c**The dimensionless E-T model achieved a maximum accuracy of 72%, with a higher printable index of 22.2%. LOF: lack-of-fusion. KH: Keyholing. Ball: Balling. Red dotted line: Keyholing boundary line. Dashed black line: Maximum hatch spacing needed to fabricate a fully dense part. Circle: Experimentally observed defect-free single track. Square: Ex-obs lack-of-fusion defect. Triangle: Experimentally observed balling defect. Diamond: Experimentally observed keyholing defect. Salmon-shaded region: Predicted lack-of-fusion region. Blue-shaded region: Predicted keyholing region. Teal-shaded region: Predicted balling region.

70 $\mu$m, 80 $\mu$m, and 30 $\mu$m, respectively. Although various studies have reported a wide range of process conditions, these selected values effectively represent the central trends observed in the literature.

The analytical E-T model and the NN (ML) E-T model achieved the highest accuracy of 76%, as Fig. 5a shows. Figure 5 displays the printability maps with the highest accuracy for each thermal model; however, the Supplementary Material includes all twelve printability maps for each model. Using the criteria set Equations (12), (17), and (14) (**LOF2-KH2-Ball1**, see Table 7 for label details), we determined the values for the highest accuracy for their respective defects. Furthermore, we used the Python

**Table 1 | We employed a supercomputer at the HPRC facility at Texas A&M University for parallelized computational tasks**

| System | Details |
|---|---|
| System | High-Performance Research Computing (HPRC) facility at Texas A&M University |
| CPU Configuration | 1 node with 8 CPUs per node |
| Memory | 100 GB RAM |

The CPU configuration included one node with eight CPUs per node and 100 GB of requested RAM.

packages OpenCV and Sci-kit Image to analyze the relative size of the printable region (i.e., the percentage of the map that is white). The *printability index* refers to the area of the printable region. The printability maps with the highest accuracy, generated using the analytical E-T thermal model, had a printable index of 17.1%. The NN (ML) E-T thermal model with the highest accuracy had a printable index of 18.5%. Conversely, the dimensionless E-T model achieved the highest accuracy of 72% and a printable index of 22.2% using the following set of criteria: Equation (11) for lack-of-fusion, Equation (16) for keyholing, and Equation (14) for balling (**LOF1-KH1-Ball1**).

As noted above, we fixed the hatch spacing, beam diameter, and powder layer thickness to make our analysis tractable. Consequently, the accuracies of different criteria sets may be slightly underestimated. For completeness, the printability maps show the beam diameter and powder layer thickness for each experimental point, where 0 indicates a particular parameter. The Supplementary Material contains the values for precision and recall. For many misclassifications associated with balling and keyholing, the experimental points lie at the boundary of the respective defect and the neighboring regions. Therefore, minor inaccuracies during the evaluation of the criteria set made the predictions more sensitive to these boundary conditions. However, these inaccuracies did not affect the accuracy that determined the ranking of the printability maps. For the analytical and NN (ML) E-T model, the model-criteria performance achieved similar results, with the highest accuracy for both reaching 76% and utilizing the same combination of optimal criteria. We benchmarked the thermal models using the High-Performance Research Computing (HPRC) facility at Texas A&M University. Table 1 summarizes the hardware specifications.

Generating the results required approximately 1.5 hours for the analytical E-T model in a P-V space of 17-by-14. In contrast, the computational expense for the NN (ML) E-T model completed computations in 3 to 5 seconds in the same P-V space, delivering a nearly *1,000-fold* acceleration with almost no cost in accuracy relative to the analytical E-T model. The dimensionless E-T model demonstrated lower accuracy due to the reduced sampling of the processing space, which resulted in excluding approximately 10 experimental points from the evaluation of printability maps compared to the other two E-T models. Table 2 details the computational expense and parallelization capability of each method for a single composition exploring a 17-by-14 P-V space, with model timings that may vary based on the resolution of the processing space.

Previous work by the authors[19] demonstrated that generating a printability map calibrated against experiments requires approximately 10 days for a single composition. This timeframe includes sample preparation and the execution of experiments to evaluate melt pool characteristics as well as process-induced defects, such as lack-of-fusion, balling, and keyholing. Although a precise breakdown of the time required for each step is not available, the cumulative duration highlights the challenges associated with experimental screening, particularly for high-throughput applications. In contrast, computational methods, such as the NN (ML) E-T model, can evaluate a composition in just 3 to 5 seconds, offering a practical and efficient alternative for large-scale compositional exploration.

Moreover, for the dimensionless form of the E-T model, Fig. 5c shows smoother boundary regions compared to Fig. 5a and Fig. 5b. The dimensionless E-T model describes reality more accurately with its smooth boundaries than the staggered boundaries obtained from the analytical E-T and NN (ML) E-T models. Concurrently, the dimensionless E-T model matches the NN (ML) E-T model in computational expense, requiring only 3 to 5 seconds. As stated above, the equiatomic CoCrFeMnNi alloy ranks as the most well-researched single-phase FCC HEA. Leveraging this alloy establishes a 'ground truth' for HEAs that exhibit a single-phase FCC structure at elevated temperatures with elements Co, Cr, Fe, Mn, and Ni. Ultimately, the predictions identify Equations (12), (17), and (14) (**LOF2-KH2-Ball1**, see Table 7 for label description) as the optimal combination criteria for a general CoCrFeMnNi alloy.

### Ensemble probability printability map for CoCrFeMnNi high entropy alloy

After constructing probability maps for the various combinations of criteria listed in Table 7, we aggregated the resulting 12 printability maps into a unified probabilistic representation, as detailed in **Ensemble Probability Printability Map**. Figure 6 depicts the ensemble probabilistic printability maps, which illustrate the likelihood of defects occurring across the power-velocity domain.

Notably, for the Cantor alloy system, the printable region (with 0% probability of defects) appears narrow, yet the central region of the map shows significantly lower probabilities of defects. This trend arises from a consensus among models indicating a defect-free zone. Moreover, regions where models concur on the presence of specific defects exhibit heightened probabilities of occurrence. For instance, in the bottom right corner, all models agree on parameters that lead to lack-of-fusion, increasing its probability for those power-velocity combinations. Among the criteria sets, **LOF2-KH2-Ball1** (see Table 7 for label details) stands out with the highest average model probability (19%), as Table 3 delineates. Across all three maps in Fig. 6, the models agree on areas prone to lack-of-fusion, keyholing, and balling. Additionally, the variance in accuracy values among the 12 printability maps, ranging from 3% to 4%, indicates a consistent predictive performance across the three thermal models. This consistency suggests a reliable level of agreement among the models for a given alloy. These ensemble printability maps offer a means to assess uncertainty at each power-velocity point, offering valuable insights into the reliability of predictions.

### Alloy design for printability and high temperature applications

The previous analysis identifies an alloy's processing window by randomly selecting it from the vast Co-Cr-Fe-Mn-Ni alloy space. Through *co-design*, the process optimizes alloys for high-temperature applications and amenability to AM. In order to design Cantor alloys for use in high-temperature applications, we screened the system for alloys predicted to exhibit a single FCC phase at 1300°C using the Thermo-Calc TCHEA5 database for high-entropy alloys. To design printable alloys, one must rely on composition-based alloy design metrics. Such composition-based objectives are the hot cracking susceptibility coefficient, the $\tau_{solid}/\tau_{spread}$ balling indicator, and the solidification range $\Delta T$. We calculated the phase stability constraint and the three composition-based printability metrics for a subset of quinary alloys within the Cantor system. We discretized the compositional simplex space with 5 at% spacing, resulting in 20 divisions per dimension, which resulted in a total of 3876 alloys (ignoring points that did not include all the elements at once).

We calculated this number using Equation (1):

$$N_{\text{internal}} = \binom{n+d-1}{d-1} - \binom{n+d-2}{d-2}, \tag{1}$$

where $n$ is the number of divisions per dimension (20 for 5 at% spacing) and $d$ is the number of components (5 for quinary alloys). For $n = 20$ and $d = 5$, this calculation gives:

$$N_{\text{internal}} = \binom{24}{4} - \binom{23}{3} = 10,626 - 6,750 = 3,876. \tag{2}$$

**Table 2 | The analytical E-T thermal model took about 1.5 hours to calculate the melt pool dimensions, while the NN (ML) E-T model and the dimensionless form of the E-T model took 3 to 5 seconds for a P-V space of 17-by-14**

| Method | Time (hh:mm:ss) | Parallelization Capability |
|---|---|---|
| Analytical Thermal Model | 01:30:00 | Can be parallelized for batch jobs |
| NN (ML) E-T Model | 00:00:05 (5 seconds) | Can be parallelized for batch jobs |
| Dimensionless E-T Model | 00:00:05 (5 seconds) | Can be parallelized for batch jobs |
| Experimental Method | 10 days[19] | Linear |

The timings of the E-T models may change depending on the resolution of the explored processing space. Sheikh et al.[19] discussed that creating a printability map experimentally takes about 10 days.

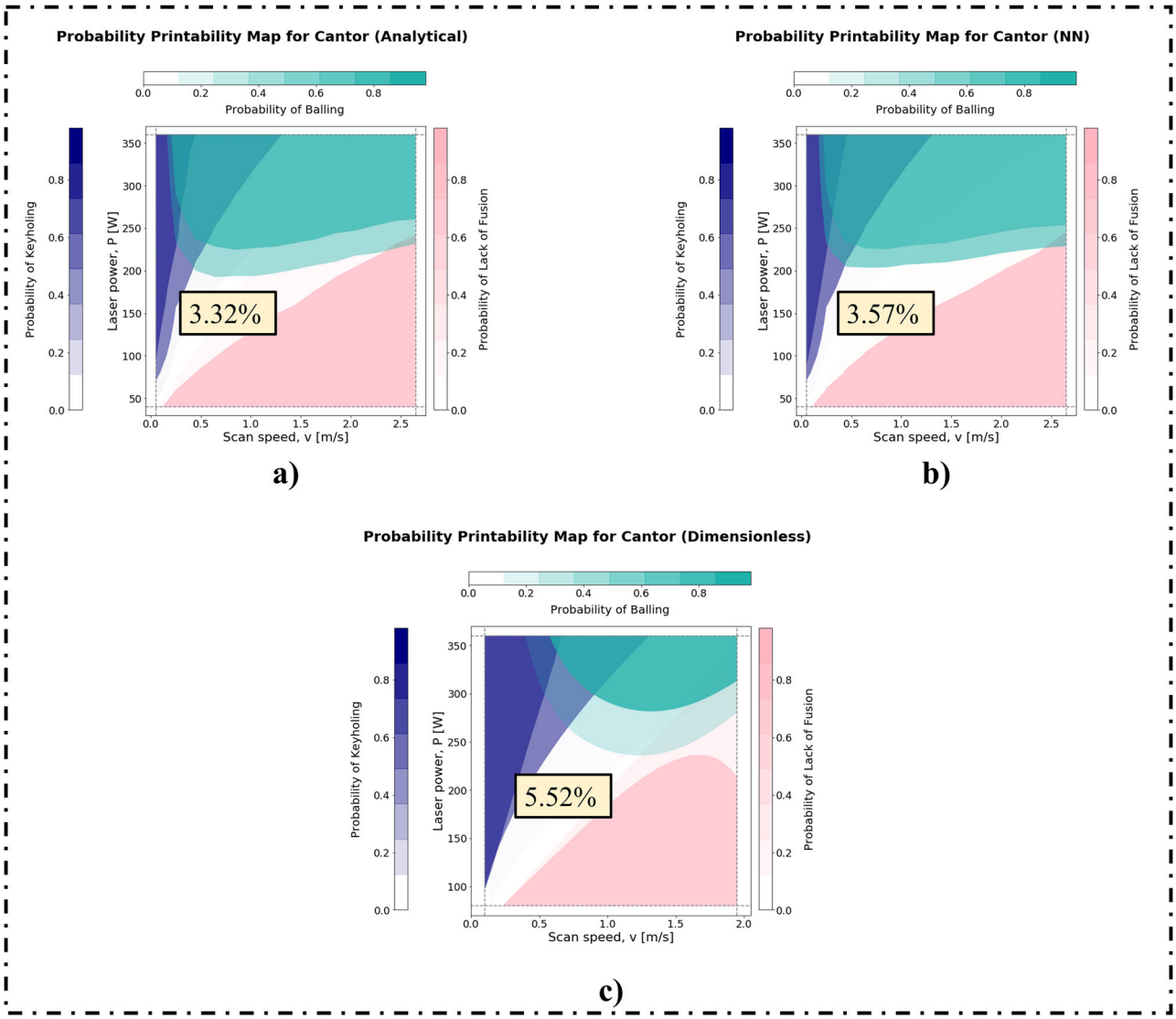



**Fig. 6 | Probability Printability Maps for Cantor Alloys.** Each of the 12 printability maps illustrates the distinction based on the thermal model type implemented. **a** The probability printability map uses the analytical E-T model melt pool dimensions, which **Analytical Eagar-Tsai Thermal Model** describes, **b** The probability printability map utilizes the NN E-T model melt pool dimensions, which **Machine Learning Model for Eagar-Tsai Thermal Model** describes, **c** The probability printability map uses the dimensionless form of the E-T model melt pool dimensions, which **Eagar-Tsai Thermal Model in Dimensionless Form** describes. These maps address the three decision problems for each defect criterion related to defect presence (i.e., lack-of-fusion, keyholing, and balling). The transparency of the region determines the likelihood of a defect occurring within the power-velocity space.

Equation (1) accounts for all points in the simplex form (including the boundary points) and subtracts the bound points to obtain the internal points (all 5 elements with a minimum of 5 at% present). This discretization approach provides a representative yet computationally feasible subset of the continuous chemical space. Figure 7a depicts in red the 58.46% of the space failed the HTP phase stability constraint.

Figure 7a shows the Pareto-front of optimal alloys for AM in blue and identifies them within the feasible space. In addition, Fig. 7b, c, and d show the 2-dimensional projections of this 3-dimensional objective space, with color depicting the third objective. To visualize how printability metrics vary within the composition space, this work employs a dimensional reduction technique called t-distributed stochastic neighbor embedding (t-SNE) to

**Table 3 | The accuracy and model probability across the three different E-T models provide a way to compare them**

| Type of E-T Model: | Analytical | | ML Model | | Dimensionless | | Avg. Mod. Prob. |
|---|---|---|---|---|---|---|---|
| | Acc. | Mod. Prob. | Acc. | Mod. Prob. | Acc. | Mod. Prob. | |
| **LOF1-KH1-Ball1** | 70% | 8% | 70% | 8% | 72% | 12% | **10%** |
| **LOF1-KH1-Ball2** | 68% | 6% | 68% | 6% | 71% | 10% | **7%** |
| **LOF1-KH2-Ball1** | 66% | 4% | 70% | 8% | 68% | 6% | **6%** |
| **LOF1-KH2-Ball2** | 69% | 7% | 67% | 5% | 68% | 6% | **6%** |
| **LOF1-KH3-Ball1** | 65% | 3% | 65% | 3% | 61% | 1% | **2%** |
| **LOF1-KH3-Ball2** | 64% | 2% | 63% | 2% | 62% | 2% | **2%** |
| **LOF2-KH1-Ball1** | 72% | 12% | 74% | 18% | 68% | 6% | **12%** |
| **LOF2-KH1-Ball2** | 69% | 7% | 75% | 22% | 67% | 5% | **11%** |
| **LOF2-KH2-Ball1** | 84% | 10% | 87% | 14% | 91% | 21% | **19%** |
| **LOF2-KH2-Ball2** | 76% | 27% | 76% | 27% | 66% | 4% | **10%** |
| **LOF2-KH3-Ball1** | 71% | 10% | 74% | 18% | 65% | 3% | **5%** |
| **LOF2-KH3-Ball2** | 70% | 8% | 73% | 16% | 64% | 2% | **4%** |

Equation (27) calculates the model probability, while Equation (26) determines the accuracy. The average model probabilities for **LOF2-KH1-Ball1** and **LOF2-KH2-Ball1** are 12% and 19%. Table 7 in **Summary of Processing-Informed Criteria** details the shorthand for the combination of the defect labels used in column 1. *LOF* lack-of-fusion, *KH* keyholing, *Ball* balling.
The model values highlight the criteria combination and the average model probability over the three E-T models (items in bold).

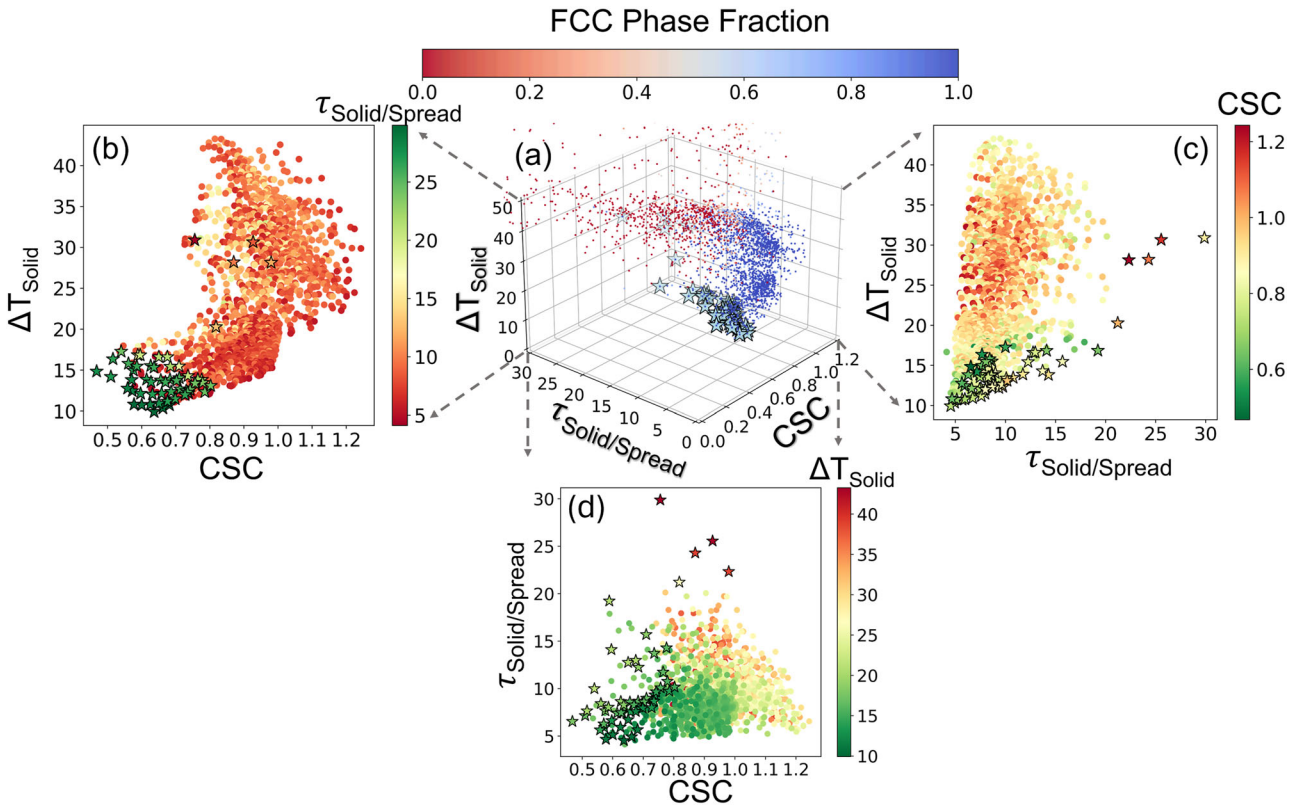



**Fig. 7 | The Tri-Objective Pareto Front of the Co-Cr-Fe-Mn-Ni Alloy Space. a** The figure plots the tri-objective Pareto front in the objective space, representing the FCC constraint as a color axis. **b** The figure maps the Pareto-front onto the $\Delta T_{Solid}$-CSC axis with $\tau_{solid}/\tau_{spread}$ as the color axis. **c** The Pareto-front projected onto the $\tau_{solid}/\tau_{spread}$-CSC plane with $\Delta T_{Solid}$ as the color axis. **d** The figure projects the Pareto-front on the $\Delta T_{Solid}$-$\tau_{solid}/\tau_{spread}$ axis with CSC as the color axis. CSC hot cracking susceptibility coefficient. $\tau_{solid}$: characteristic solidification time. $\tau_{spread}$: characteristic spreading time. $\Delta T_{solid}$: solidification range. Star: Composition points that fulfill the tri-objective criteria.

project the 5-dimensional Cantor alloy space into a 2-dimensional embedding, which Fig. 8a, b illustrate. Using t-SNE, the projection places points near each other in a high-dimensional space close together in the 2-dimensional embedding. However, the Euclidean distance in the 2-dimensional embedding does not allow for quantitative inferences about the relationship between points. Each point in the t-SNE embedding represents an alloy with a unique composition. The arrangement places points near the corners of the "pentagonal" shape closer to unary alloys. For example, the alloy in the Cr-corner of the t-SNE corresponds to the unary (i.e., pure) Cr, while the edge of the "pentagonal" shape connecting Cr to Mn

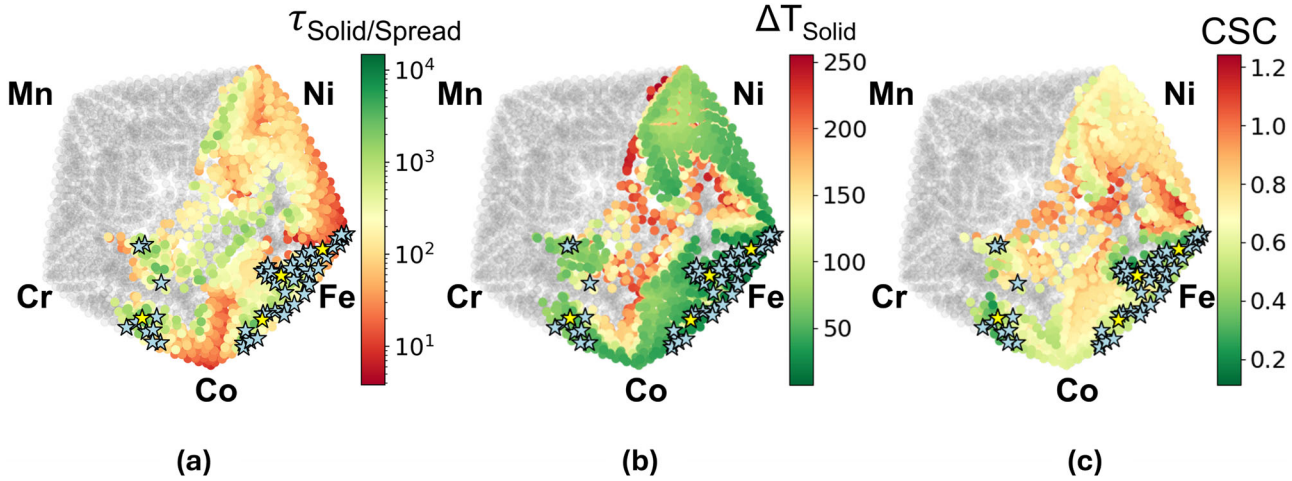



**Fig. 8 | Tri-Objective Pareto Front in the Composition Space for the Co-Cr-Fe-Mn-Ni Alloy Space.** The tri-objective Pareto front plots in the composition space. Grey represents the alloys that fail the FCC constraint. The t-SNE projections of the CoCrFeMnNi alloy space display **a** the ratio of characteristic solidification time to characteristic spreading time, **b** the soldification range and **c** the cracking coefficient, highlighting how the printability objectives vary with composition. CSC hot cracking susceptibility coefficient. $\tau_{solid}$: characteristic solidification time. $\tau_{spread}$: characteristic spreading time. $\Delta T_{solid}$: solidification range. Star: Composition points that fulfill the tri-objective criteria.

**Table 4 | We constructed the ensemble probability printability maps for the five sampled alloys using three E-T thermal models**

| Alloy | Composition | Analytical E-T Model Ranking | NN (ML) E-T Model Ranking | Dimensionless E-T Model Ranking |
|---|---|---|---|---|
| Alloy A | $Co_{15}Cr_{15}Fe_{60}Mn_5Ni_5$ | 5 | 5 | 5 |
| Alloy B | $Co_{45}Cr_{20}Fe_{25}Mn_5Ni_5$ | 3 | 3 | 3 |
| Alloy C | $Co_{30}Cr_{30}Fe_{40}Mn_5Ni_5$ | 4 | 4 | 4 |
| Alloy D | $Co_{40}Cr_{30}Fe_{20}Mn_5Ni_5$ | 2 | 2 | 2 |
| Alloy E | $Co_{20}Cr_{40}Fe_5Mn_5Ni_{30}$ | 1 | 1 | 1 |
| **Computational Cost (hh:mm:ss)** | – | 01:30:00 | 00:00:05 | 00:00:05 |

We calculated the printability index for each map and ranked the alloys from the highest printability index (1) to the lowest (5). The E-T thermal models consistently ranked the five alloys in the same order, with Alloy E achieving the highest printability index and Alloy A the lowest. The table highlights the computational cost required to construct each map to highlight the associated expense.

features binary alloys consisting of Cr-Mn. Upon inspection of Fig. 8, it becomes evident that regions rich in FCC-forming elements Mn and Cr do not pass the phase stability constraint, which the t-SNE projections depict as gray.

Regions rich in Co-Fe and Cr-Co achieve the optimal trade-off between balling resistance and hot cracking resistance. A k-medoids-based sampling scheme groups the Pareto front into five representative clusters based on composition. The proposed framework analyzes the printability of these medoids (the representative members of each of the clusters, depicted as yellow stars) and queries their printability indices.

### Probabilistic printability maps for the selected alloys

After analyzing the tri-objective Pareto front and applying a k-medoids sampling, the proposed framework analyzed five alloys using the ensemble probability printability maps. The five sampled alloys include:

1. $Co_{15}Cr_{15}Fe_{60}Mn_5Ni_5$**(Alloy A)**
2. $Co_{45}Cr_{20}Fe_{25}Mn_5Ni_5$**(Alloy B)**
3. $Co_{30}Cr_{30}Fe_{40}Mn_5Ni_5$**(Alloy C)**
4. $Co_{40}Cr_{30}Fe_{20}Mn_5Ni_5$**(Alloy D)**
5. $Co_{20}Cr_{40}Fe_5Mn_5Ni_{30}$**(Alloy E)**.

As before, the printability analysis utilized a 17-by-14 design of experiments, varying power and velocity while keeping the hatch spacing, laser beam diameter, and powder layer thickness constant at 70 $\mu$m, 80 $\mu$m, and 30 $\mu$m, respectively. These conditions remained consistent with those applied to the equiatomic CoCrFeMnNi alloy. Using the melt pool profiles calculated from the three thermal E-T models, we generated ensemble probability printability maps for the Pareto front alloys. These maps provide a robust assessment of alloy printability by incorporating the uncertainty associated with each defect criterion combination listed in Table 7. We applied the methodology described in **Ensemble Probability Printability Map**, along with the model probabilities calculated in **Ensemble Probability Printability Map for CoCrFeMnNi High Entropy Alloy**, to construct ensemble probability printability maps for five sampled alloys within the Pareto set. This approach assumes that the accuracies, derived by comparing predictive printability maps to experimentally determined outcomes for the Cantor system, transfer to other alloys within the same chemical space. Based on these maps, we ranked the alloys by their printability index and summarized the results in Table 4.

The ensemble probability printability maps generated using the NN (ML) E-T thermal model, illustrated in Fig. 9, showed that the alloy that had a maximum printability index was **Alloy E**, with a maximum printability index of 5.89%, followed by the remaining alloys: **Alloy D** (Fig. 9d), **Alloy B** (Fig. 9b), **Alloy C** (Fig. 9c), and **Alloy A** (Fig. 9a) with a minimum printability index of 4.22%, as summarized in Table 4. The NN (ML) E-T thermal model achieved a higher maximum printability index of 5.89% when compared to the ensemble probability printability map constructed for the equiatomic CoCrFeMnNi alloy using the NN (ML) E-T thermal model. In addition, Fig. 9 compares the printability indices of the ensemble probability printability maps to the ensemble

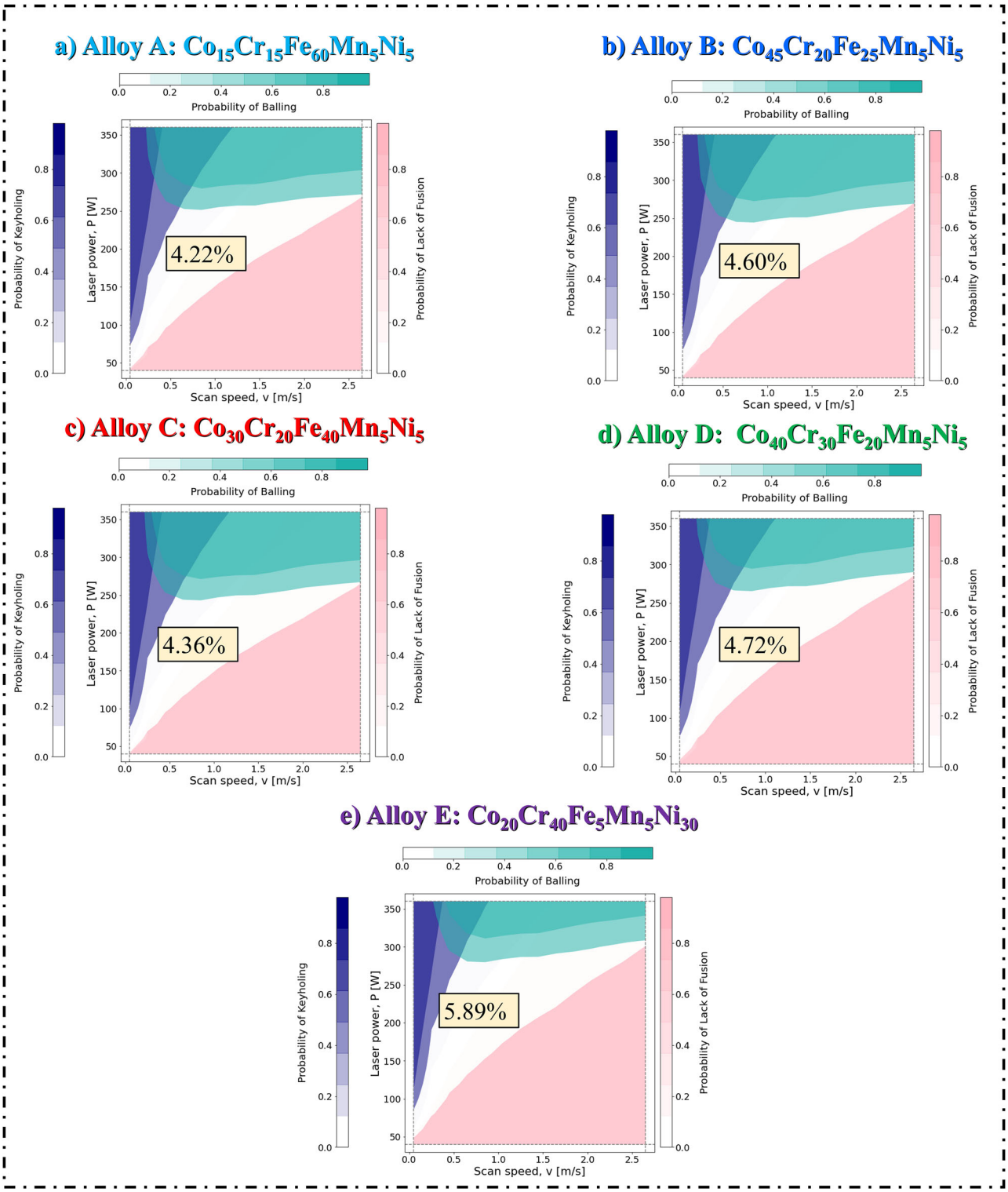



**Fig. 9 | Ensemble probability printability maps for pareto front alloys using NN E-T Model.** Using the NN (ML) E-T model to calculate the melt pool profile for the five sampled alloys. The printability map for **a** Alloy A, **b** Alloy B, **c** Alloy C, **d** Alloy D and **e** Alloy E are displayed. **Alloy E** ($Co_{20}Cr_{40}Mn_5Fe_5Ni_{30}$) achieved the highest printability index of 5.89%. **Alloy A** ($Co_{15}Cr_{15}Mn_5Fe_{60}Ni_5$) has the lowest printability index at 4.22%. Both printability index values exceed the printability index of 3.52%, calculated from the printability maps generated using the NN (ML) E-T model for the equiatomic CoCrFeMnNi. Salmon-shaded region: Area that represents the predicted lack-of-fusion region. Blue-shaded region: Area that represents the predicted keyholing region. Teal-shaded region: Area that represents the predicted balling region. Red dashed line: Keyholing boundary. Black dashed lines: Maximum hatch spacing required for sufficient overlap in a fully built part.

probability printability maps generated using the analytical E-T thermal model for the Pareto front alloys and presented in the Supplementary Material. For the analytical E-T model, the maximum printability index was 5.62% for **Alloy E**, while the minimum printability index was 4.12% for **Alloy A**. The difference in the values for the indices calculated for the NN E-T model and the analytical E-T model is less than 2%. The NN (ML) E-T thermal model demonstrated higher computational efficiency, producing results approximately **1000 times faster** than the analytical

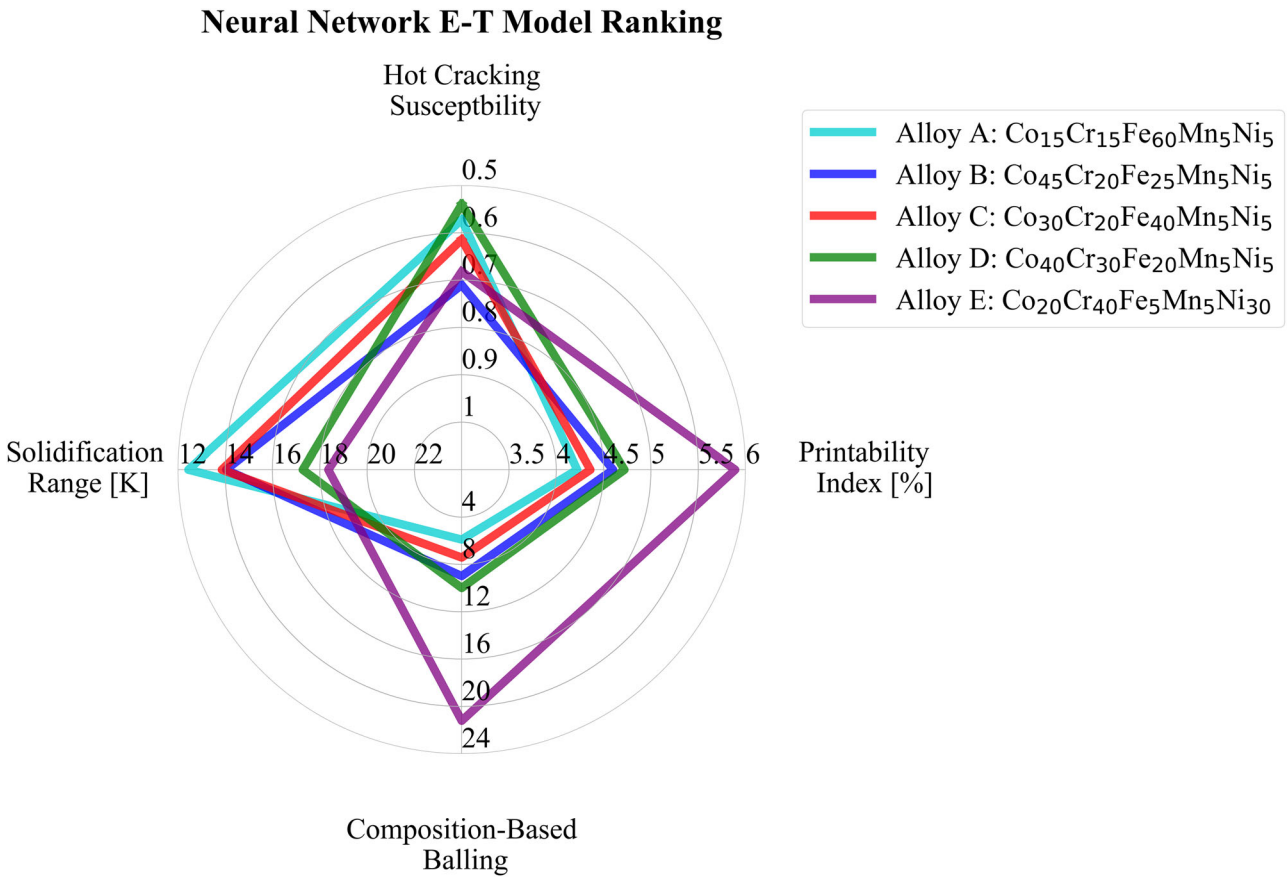



**Fig. 10 | Ranking optimized alloys for high-temperature application with NN E-T model.** The radar chart summarizes the results for the alloys predicted using the Pareto front objectives and further analyzes the performance and printability of the alloys based on the four objectives: hot cracking susceptibility, solidification range, composition-based balling, and the printability index when using the neural network E-T model. The chart inverts the axis for hot cracking susceptibility and solidification range to maximize all objectives.

E-T model, enabling the high-throughput assessment of printability over large chemical spaces.

Figure 10 graphically summarizes the ranking of the alloys using the neural network surrogate E-T model. Figure 10, the radar chart inverts the axes for hot cracking susceptibility and solidification range to the problems as one of maximization. We want to note that the difference in printability index value between ensemble probability printability maps is 0.12% and 1.67% for the NN surrogate model.

The high computational efficiency and predictive accuracy of the NN (ML) surrogate for the E-T model make it suitable for HTP applications. In HTP workflows, where the exploration of vast compositional and parameter spaces is essential, the NN surrogate model enables rapid evaluation of alloy printability with considerable computational speedup when compared to the analytical E-T model. This acceleration significantly reduces the time required to identify optimal alloy compositions and processing conditions. Additionally, the ensemble probabilistic framework enhances the robustness of the HTP methodology by integrating multiple defect criteria and accounting for uncertainty through weighted evaluations. This approach provides a more comprehensive assessment of printability, ensuring confidence in the rankings even when empirical accuracy data for specific criteria are unavailable. By leveraging both single-criteria and ensemble-based approaches, the framework offers flexibility to accommodate different stages of the HTP workflow, from rapid initial screening to detailed final evaluation.

Another method to assess the printability of the Pareto front alloys is to use a single criterion set, such as the optimal criteria set of **LOF2-KH2-Ball1**, to rank the alloys. While using a single criteria set may not provide a full assessment of alloy printability, it can help reinforce confidence in the rankings. The Supplementary Material presents the analysis of the ranking of the Pareto front alloys based on the optimal criteria, where **Alloy E** emerged as the most printable alloy, agreeing with the ensemble probability printability maps results. This consistency across both the ensemble and single-criteria approaches reinforces confidence in the rankings and demonstrates the robustness of the proposed framework. The ensemble probabilistic printability map approach provides a more comprehensive analysis. By weighting individual criteria sets based on their assessed accuracies, which result from comparing predictions against empirical data, the ensemble framework incorporates uncertainty into the rankings. When accuracy data for specific criteria sets is unavailable, a conservative strategy assigns equal weights. The scalability, adaptability, and efficiency of the proposed framework make it a powerful tool for accelerating material discovery and optimization in L-PBF.

## Discussion

In **Results**, we designed five alloys with composition-based printability objectives to minimize balling and solidification cracking. *However, there exist opportunities to tailor alloys with optimal printability maps*. From our previous analysis, 1610 alloys satisfied the 1300 °C single-phase FCC constraint. The ET-NN model and the optimal criteria combination determined in **Printability of an Arbitrary Co-Cr-Fe-Mn-Ni High Entropy Alloy System** the intrinsic printability of these 1610 alloys in an HTP manner. Specifically, we queried the printability maps for the 1610 constraint-satisfying alloys, and the printability index was extracted from each printability map. As defined above, the printability index represents the defect-free area in a given processing range. Using this index as an alloy design metric, the processable windows ranked the alloys. Figure 11 reveals

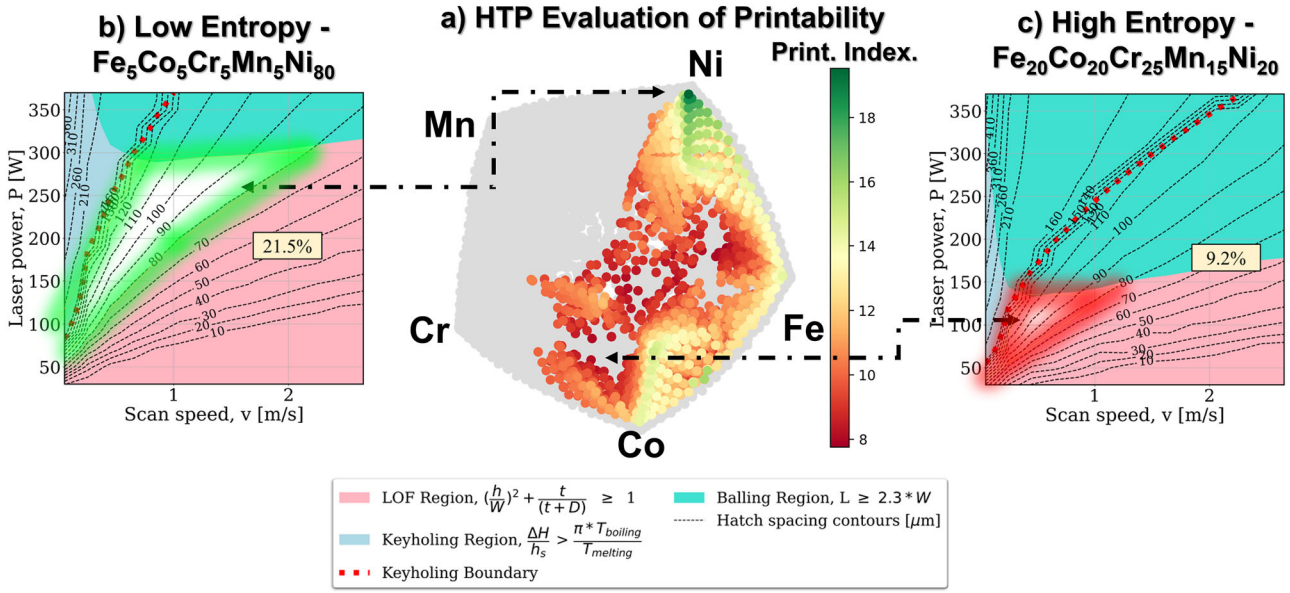



**Fig. 11 | Printability Index comparison between a low entropy alloy and a high entropy alloy. a** A t-SNE projection visualizes design space, with the printability index plotted over this space. Comparing **b** high entropy alloy and **c** low entropy alloy printability maps reveals that the low entropy alloy achieves a higher printability index because defects such as balling and keyholing are less prominent compared to the high entropy alloy. The low entropy alloy achieves a printability index of 21.5%. On the other hand, the high entropy alloy achieves a printability index of 9.2%. Print. Index Printability Index, W melt pool width, h hatch spacing, t powder layer thickness, D melt pool width. ΔH: specific enthalpy. $T_{boiling}$: boiling temperature. $T_{melting}$: melting temperature. $h_s$: specific enthalpy at melting. Salmon-shaded region: Area that represents the predicted lack-of-fusion region. Blue-shaded region: area that represents the predicted keyholing region. Teal-shaded region: area that represents the predicted balling region. Red dashed line: keyholing boundary. Black dashed lines: Maximum hatch spacing required for sufficient overlap in a fully built part.

that Ni-rich alloys are the *most intrinsically printable alloys* within the Co-Cr-Fe-Mn-Ni alloy space. Additionally, Fig. 11 highlights that alloys with *higher compositional entropy*, i.e., those in the central regions of the composition space, *are less printable*.

This work relies on a curated dataset derived from an in-house database of L-PBF alloys. We filtered the dataset using objective criteria described in **Results** to align with the specific requirements of L-PBF applications. The filtering process focused on experimental data with detailed process parameters (e.g., power, velocity), defect outcomes (e.g., balling, keyholing, lack of fusion), and build-state characteristics. The Cantor alloy system (Co-Cr-Fe-Mn-Ni) and its derivatives naturally emerged as prominent components due to their extensive characterization in additive manufacturing literature and the availability of robust experimental data. While broader datasets, such as Citrine/UCSB MPEA[70] or ULTERA[71], provide valuable insights into general high-entropy alloy systems, they lack the detailed process-parameter-dependent defect data required for the objectives of this study. By using a dataset specifically tailored to L-PBF, the findings remain focused and directly relevant to the development of defect prediction frameworks for this manufacturing method.

The most prominent difference between the high and low entropy alloys lies in the size of the balling defect region in the printability maps, as shown in Fig. 11b, c. Equation (14) defines criterion used to determine balling in this printability map, based on the ratio between melt pool length and width. The difference in L/W ratios between these two alloys stems from increased alloy complexity, which increases phonon and electron scattering and subsequently decreases the thermal conductivity of the alloy[72].

The Cantor alloy with a higher entropy has a predicted thermal conductivity of 21 *W/m/K*, while the Cantor alloy with a lower entropy has a predicted thermal conductivity of 38 *W/m/K*. Thermal conductivity directly impacts the melt pool length. As the thermal conductivity decreases, the time required for the melt pool to solidify increases ($4.66 \times 10^{-4}$ s for the Cantor alloy with lower entropy and $6.58 \times 10^{-4}$ s for the Cantor alloy with higher entropy, as predicted with Equation (21)). This increase in solidification time produces a long melt pool of $4.38 \times 10^{-4}$ m for the Cantor alloy with higher entropy compared to $2.58 \times 10^{-4}$ m for the Cantor alloy with lower entropy (given $P = 250$ W and $v = 1.45$ *m/s*). This longer liquid melt pool becomes more susceptible to Rayleigh capillary instability. Although thermal conductivity affects the melt pool width, its impact is less pronounced compared to the melt pool length. Consequently, (given $P = 250$ W and $v = 1.45$ *m/s*), the Length (L) /Width (W) ratio of the Cantor alloy with higher entropy (L/W = 3.65) exceeds that of the Cantor alloy with lower entropy (L/W = 2.09). This higher L/W ratio increases susceptibility to balling in the Cantor alloy with higher entropy.

Alloy design relies on *optimizable* indicators of quantities of interest associated with the alloy. Designing alloys amenability to AM requires easily queriable metrics to indicate the size of the processing parameter window that produces defect-free prints. Just as Vela et al.[73] demonstrated an indicator correlated with the extent of the processing window that prevents balling, alloy design schemes must *maximize* the printability index to yield alloys with a wide range of processing parameters that result in good prints. Combining the printability index with indicators that describe the tendency toward hot-cracking (i.e., solidification range and CSC) creates opportunities to design *intrinsically printable alloys*, i.e., alloys resistant to processing-related porosity and solidification-related cracking. In addition, the probabilistic framework illustrated in Fig. 6 for the equiatomic Cantor alloy and Fig. 9 for the sampled alloys highlight the value of these tools for alloy design and process optimization.

The proposed framework screens the printability of 1000s of candidate alloys in an HTP manner. This framework enables the HTP screening of alloys that are resistant to porosity defects. This will allow for better allocation of resources for experimental investigation. The observed trend of reduced printability in more "high entropy" alloys highlights a need for targeted alloying strategies to balance compositional complexity and processability. A trade-off exists between compositional complexity and printability. By identifying Ni-rich alloys as more printable alloys, the study

provides actionable insights into the development of L-PBF material in the Co-Cr-Fe-Mn-Ni system. Moreover, the probabilistic printability maps developed in this work provide defect predictions with quantified uncertainty. Uncertainty quantification during AM plays a critical role in applications that require ensuring quality assurance.

## Methods

### Calculation of thermophysical properties

We used several models to calculate the thermo-physical properties across the Cantor alloy system. The Property, Equilibrium, and Scheil CALPHAD models in the Thermo-Calc software derive the thermo-physical properties using thermodynamic databases TCHEA5, TCAL7, and TCNI11[74,75]. These models compute various material properties relevant to the L-PBF process, such as the solidus and liquidus temperatures, and the latent heat of fusion. Furthermore, at room temperature (298 K), solidus temperature, and liquidus temperature, the software queries the following material properties: density, electric conductivity, heat capacity, thermal conductivity, thermal diffusivity, thermal resistivity, enthalpy, dynamic viscosity, and kinematic viscosity. Besides CALPHAD-based models, the rule-of-mixtures models (ROMs) calculate the compositionally weight average of the melting temperature, boiling temperature, density, and molecular weight. The averaging is given by $P_{alloy} = \Sigma_i^n x_i P_i$-where $i$ represents the set of atomic species, $x_i$ is the fraction of the $i$-$th$ element in the alloy value, and $P$ is the material property of the $i$th element[76,77]. The Python Materials Genomics library (`pymatgen`)[78], a Python library, provides the elemental properties. The CALPHAD-derived and ROM-derived properties serve as inputs to thermal models and variables in AM-related dimensionless numbers, as discussed in **Analytical Eagar-Tsai Thermal Model**, **Eagar-Tsai Thermal Model in Dimensionless Form** and **Machine Learning Model for Eagar-Tsai Thermal Model**.

The CALPHAD-based material properties and the ROM-calculated material properties calculate additional properties. For example, $T_{\text{liquidus}} - T_{\text{solidus}}$ defines the solidification range, where $T_{\text{liquidus}}$ is the liquidus temperature and $T_{\text{solidus}}$ is the solidus temperature[74]. Moreover, $H_{\text{liquidus}} - H_{\text{RT}}$ determines the total enthalpy, where $H_{\text{liquidus}}$ represents the enthalpy at liquidus, and $H_{\text{RT}}$ represents the enthalpy at room temperature[79,80]. Using the total enthalpy, $H_{\text{total}}/(T_{\text{liquidus}} - T_{\text{RT}})$ describes the effective heat capacity, where $T_{\text{RT}}$ is the room temperature[81]. Similarly, $H_{\text{liquidus}} - H_{\text{solidus}}$ quantifies the melting enthalpy, and $10 \times R \times T_{\text{liquidus}}$ indicates the boiling enthalpy, where $H_{\text{solidus}}$ is the enthalpy at the solidus temperature and $R$ is the gas constant[74,79]. Furthermore, $H_{\text{liquidus}} + C_{p,\text{liquidus}} \times (T_{\text{boiling}} - T_{\text{liquidus}})$ calculates the enthalpy at boiling, and $H_{\text{at-boiling}} + H_{\text{boiling}}$ describes the enthalpy after boiling. In this, $C_{p,\text{liquidus}}$ represents the specific heat at the liquidus temperature, $T_{\text{boiling}}$ is the boiling temperature, $H_{\text{at-boiling}}$ represents the enthalpy at boiling, and $H_{\text{boiling}}$ represents the boiling enthalpy[82–84].

Furthermore, a composition-based feature vector (CBFV) provides further insight into the material system. A CBFV uses a data-driven approach to calculate physically interpretable results about the material system[85]. CBFV contains descriptive statistics such as the average, range, sum, and variance of the constituent elements. The package features the alloy system using the Oliynik data set, which uses element chemistry to predict the stoichiometric material properties[86].

### Analytical Eagar-Tsai thermal model

Eagar-Tsai (E-T) model [22] provides an inexpensive analytical model to predict the temperature distribution for a moving heat source with a Gaussian profile on a substrate during welding. Overall, good agreement between the calculated melt pool dimensions from the E-T model and the experimental data has made the E-T model a reliable method for calculating the melt pool dimensions of single tracks within the AM community[6,15,27,87]. Green's function for the steady-state heat conduction equation expresses the substrate's temperature profile. To address the complexity of the solution involving a double integral, the model simplifies it by approximating the input heat source profile with a Gaussian function. Additionally, the model assumes a quasi-steady state semi-infinite substrate with temperature-independent material properties while ignoring convective and radiative heat flow to simplify the model. The final form of the integration used to calculate the temperature distribution is expressed in the following form:

$$T - T_0 = \frac{1}{2}\int_{t'=0}^{t'=t} dT_{t'} = \frac{q}{\pi\rho C_p\sqrt{4\pi\alpha}} \int_{t'=0}^{t'=t} \left[\frac{(t-t')^{-\frac{1}{2}}}{2\alpha(t-t')+\sigma^2} e^{\left[-\frac{(x-\nu t')^2+y^2}{4\alpha(t-t')+2\sigma^2} - \frac{z^2}{4\alpha(t-t')}\right]}\right] dt', \tag{3}$$

In the above equation, $T_0$ represents the initial temperature of the substrate, and $T$ predicts the temperature at time $t$ at any point with a coordinate of $(x,y,x)$. The E-T model categorizes its input variables into two groups: *process parameters* and *material properties*. The process parameters are $q, \mu$, and $\sigma$, where $q$ denotes the input heat power supplied to the material, $\mu$ represents the scanning speed of the moving heat source, and $\sigma$ corresponds to the standard deviation of the Gaussian laser profile. On the other hand, material properties include $\kappa$, the thermal conductivity; $\rho$, the density of the printed material; $C_p$, the specific heat capacity; and $\alpha$, the thermal diffusivity. The following equation calculates $\alpha$:

$$\alpha = \frac{\kappa}{\rho C_p} \tag{4}$$

To estimate the melt pool length, width, and depth, the numerical quadrature method calculates the temperature at each point by applying it to Equation (3). The comparison between the calculated temperature and the material's melting temperature determines the melt pool boundary.[11].

### Eagar-Tsai thermal model in dimensionless form

To enable faster and more convenient calculation of melt pool dimensions, the E-T model in Equation (3) expresses the variables $x$, $y$, and $z$ as normalized dimensionless variables $x_n = \frac{x}{a}$, $y_n = \frac{y}{a}$, and $z_n = \frac{z}{\sqrt{D/(ua)}}$, respectively[7,27]. This change of variables produces the following solution for Equation (3):

$$T = T_s g(x_n, y_n, z_n, p) = T_m B g(x_n, y_n, z_n, p) \tag{5}$$

In the above equation, $g$ represents a universal function in terms of the given coordinate and dimensionless parameter $p$, defined as the laser dwell time ($t_d = \frac{a}{\nu}$) over the thermal diffusion time ($t_D = \frac{a^2}{D}$), i.e., $p = \frac{D}{ua}$. This function captures the effect of the laser scan velocity on the temperature distribution during the AM processes. $B$ represents another dimensionless parameter defines the energy density deposited by the laser beam divided by the enthalpy of melting ($h_s = \rho C T_m$), i.e., $B = \frac{\Delta H}{2^{3/4}\pi h_s}$. $B$ captures the effect of the laser power on the temperature distribution during L-PBF.

When $p$ is smaller than 1, the diffusion time is greater than the dwell time. In other words, the beam size exceeds the thermal diffusion into depth during the dwell time, leading to a shallow and extended melt pool. This situation typically occurs when the material's thermal conductivity is low. On the other hand, a $p$ greater than 1 usually happens in materials with high thermal conductivity, increasing thermal diffusion into depth and producing a more circular shape for the melt pool. $B$ should also be much greater than 1 to have enough energy input to form a well-defined melt pool.

To estimate, the normalized melt pool length, width, and depth, the calculation must determine the melt pool boundary along each normalized coordinate axis. This corresponds to finding the spatial point on each axis where the predicted temperature in Equation (5) equals the melting temperature ($T = T_m$), i.e.:

$$Bg(x_n, y_n, z_n, p) = 1 \tag{6}$$

As a result, the melt pool dimensions can be predicted in terms of dimensionless variables $p$ and $B$ in a universal scheme. For this purpose, Rubenchik et al.[27] interpolated the melt pool dimensions calculated for

**Table 5 | The data ranges applied in the LHS design to help develop the surrogate model**

| Property | Range |
|---|---|
| Laser Speed | 0.05–5 m/s |
| Laser Power | 10–500 W |
| Beam Diameter | 50–150 $\mu$m |
| Absorptivity | 0.05–1 |
| Melting Temperature | 676–3695 °K |
| Density | 1730–19,300 Kg/$m^3$ |
| Thermal Conductivity | 10–234 W/m/°K |
| Specific Heat | 450–1400 J/Kg/°K |

**Table 6 | For the preliminary testing to determine which ML technique was necessary, the relative MSE compared the performance of the developing surrogate models**

| Method | Relative MSE |
|---|---|
| Linear Regression | 14.59 |
| Partial Least Squares | 14.60 |
| Gradient Boosting | 1.30 |
| Neural Network | 1 |

different combinations of processing parameters and materials systems. They showed that $p$ and $B$ usually change in the range of 0.1–5 and 1–20 in these calculations, respectively. These interpolated functions are:

$$L(p,B) = \frac{a}{p^2}\left[0.0053 - 0.21p + 1.3p^2 + (-0.11 - 0.17B)P^2\ln(p) + B(-0.0062 + 0.23P + 0.75p^2)\right] \quad (7)$$

$$W(p,B) = \frac{a}{Bp^3}\left[0.0021 - 0.047p + 0.34p^2 - 1.9p^3 - 0.33p^4 + B\left(0.00066 - 0.0070p - 0.00059p^2 + 2.8p^3 - 0.12p^4\right) + B^2\left(-0.00070 + 0.015p - 0.12p^2 + 0.59p^3 - 0.023p^4\right) + B^3\left(0.00001 - 0.00022p + 0.0020p^2 - 0.0085p^3 + 0.0014p^4\right)\right] \quad (8)$$

$$D(p,B) = \frac{a}{\sqrt{p}}\left[0.008 - 0.0048B - 0.047p - 0.099Bp + (0.32 + 0.015B)p\ln(p) + \ln(B)(0.0056 - 0.89p + 0.29p\ln(p))\right] \quad (9)$$

These universal algebraic functions reduce the time and complexity of calculating the melt pool dimensions in the E-T model, making it amenable for use in automated HTP materials design frameworks. Further discussion about the dimensionless E-T model and the interpolated functions for the melt pool dimension estimation appears in Rubenchik et al.[27].

### Machine learning model for Eagar-Tsai thermal model

As discussed in **Introduction**, ML models have become a popular tool in L-PBF, especially in modeling the melt pool characteristics[27–30]. Other works have also explored computational techniques such as deep learning models (DL models) to model melt pool characteristics. Hemmasian et al.[88] used Flow3D to create and train a convolutional neural network (CNN) to predict the thermal fields of the melt pool with high accuracy. Moges et al.[89] proposed a hybrid modeling framework that combines physics-based data, computational fluid dynamics (CFD) simulation data, and measurement data to train and predict the melt pool width. Raut et al.[90] introduced graph neural networks (GNNs) to model thermal distributions in L-PBF. These models offer significant benefits in computational efficiency and automate manufacturing, but they are not suited for HTP tasks.

We explore ML techniques to develop a high-fidelity surrogate for the analytical E-T model. We develop a surrogate model for E-T to circumvent its computational cost. Deploying such a predictive model requires integrating tools of physics-based analytical modeling, data-driven analysis, and calibrating it using statistical models. The framework is divided into three steps:

1. Generating a comprehensive dataset to train the ML models
2. Evaluating multiple ML techniques for their suitability in creating surrogate models, and
3. Selecting the best-performing model

The analytical E-T model generated 550,000 simulations to develop the ML surrogate model. The sample process, using the Latin hypercube (LHS) design, determined the parameters used in the simulation. Table 5displays the data ranges used for the LHS design.

We test several ML techniques to develop the surrogate model, such as Linear Regression, Partial Least Squares, Gradient Boosting, and NN. A 10 k-fold cross-validation method compared the performance of the 4 models using Mean Square Error (MSE) (defined in the Supplementary Material) as a performance metric. The MSE quantifies the relative performance of each model and is displayed in Table 6. The NN model outperformed the other models as shown in Table 6, and therefore, we chose NN as the ML technique for the surrogate model.

To develop the NN model, we normalized all inputs and outputs to have a mean of 0 and a standard deviation of 1, as normalization improves the performance of NN models[91]. The neural network (NN) includes six layers: an input layer, four fully connected hidden layers, and an output layer. The first hidden layer contains 500 units, followed by layers with 300, 250, and 150 units. The output layer has three units, predicting the melt pool's length, width, and depth. We applied the Rectified Linear Unit (ReLU) activation function to all hidden layers to introduce non-linearity. Ketkar[92] recommends using ReLU activation and the Adam optimizer in similar applications.

We trained the NN using the Adam optimizer with a learning rate of 0.001, no learning rate decay ($decay = 0.0$), $\beta_1$ set to 0.9, $\beta_2$ set to 0.999, and $\epsilon$ set to $1 \times 10^{-7}$ to prevent division by zero. We did not use the AMSGrad variant of Adam ($AMSGrad = False$). The training process lasted for 50 epochs with a batch size of 5. To validate the model's performance and ensure generalization, we implemented a 10-fold cross-validation approach. Additionally, we increased the model's complexity until it showed no further improvement in performance. This setup allowed the NN surrogate model to predict the melt pool dimensions effectively while maintaining high performance and generalization capability.

Furthermore, to improve the model's performance, we introduced a classification step that splits the results into four classes:

1. Samples that do not melt. The maximum temperature is lower than the melting temperature.
2. Melting occurs in the length and width dimensions. There is no melting in the depth dimension as the maximum temperature is close to the melting temperature.
3. Melting occurs in 3 dimensions. The melt pool depth has a maximum of 9 $\mu$m.
4. Melting occurs in 3 dimensions. The melt pool depth is larger than 9 $\mu$m.

Using the analytical E-T model, 550,000 simulations were completed in 45 days. In contrast, the NN surrogate model performed the same number of simulations in just 120 seconds. While the NN model requires an initial training phase, this is a one-time fixed cost. Once trained, the surrogate model can be deployed for large-scale simulations with drastically reduced computation time and minimal error. We intentionally designed the large training dataset to cover a broad parameter space, ensuring reliable

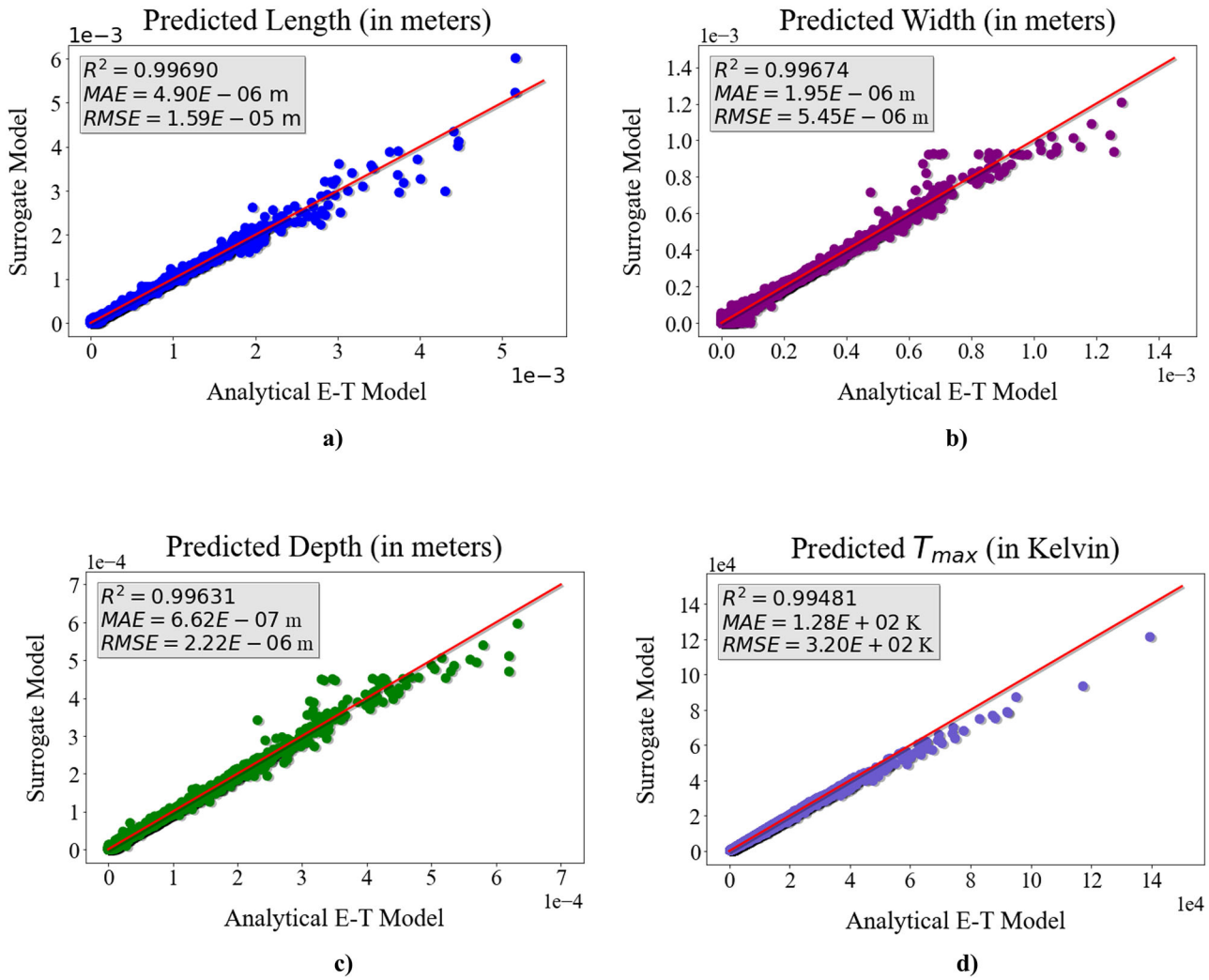



**Fig. 12 | NN Surrogate E-T Model Prediction Performance.** The analysis compares the NN surrogate model predictions to the analytical E-T model predictions. The performance metrics $R^2$, mean absolute error (MAE), and root mean squared error (RMSE) show that the NN model accurately predicts the melt pool dimensions for **a** melt pool length, **b** melt pool width, **c** melt pool depth, and **d** maximum temperature ($T_{max}$) of the melt pool at a faster computational time. K: Kelvin. m: meters.

interpolation of melt pool outputs (i.e., width, depth, and length) across various processing conditions. It is important to note that the NN surrogate model is only valid within the bounds of the processing parameters listed in Table 5, and should not be used for extrapolation beyond those ranges.

Figure 12 presents the results of the surrogate model in predicting the melt pool geometry and maximum temperature. To evaluate the NN surrogate model in predicting the melt pool dimensions, we applied three different performance metrics: root mean squared error (RMSE as defined in the Supplementary Material), mean absolute error (MAE as defined in the Supplementary Material), and coefficient of determination ($R^2$ as defined in the Supplementary Material). RMSE and MAE evaluate the predictions as the length, width, and depth can vary from units to hundreds over the parameter and composition space. In addition, $R^2$ shows how well the NN surrogate model can predict the melt pool geometries compared to the predictions of the analytical E-T model. Figure 12 illustrates that the $R^2$ value is around 0.99 for melt pool length, width, and depth, while the MAE and RMSE are in the order of $10^{-5}$-$10^{-7}$ meters for the geometry.

Furthermore, calibration can further improve the NN model's performance. However, for this study, we did not perform any calibration steps but have outlined the process in the Supplementary Material, and instead leverage the use of the pre-trained NN (ML) E-T model to estimate the melt pool dimensions for our printability maps presented.

### G-S model for the keyhole depth

Among the defect modes in AM, keyholing occurs at high laser power and low scanning speeds, i.e., too high input energy density[93]. Keyholing is characterized by the formation of pores due to the entrapment of shielding gas and a melt pool with a high depth-to-width ratio.

Therefore, analytical thermal models, such as the E-T model, fail to calculate the melt pool dimensions when keyholing occurs since they are formulated in such a way as to estimate the heat conduction mode only. Consequently, this work applies the simple expression proposed by Gladush and Smurov[24] (referred to as G-S model) to estimate the keyhole depth, as follows:

$$D_k = \frac{AP}{2\pi\kappa T_b} ln\left(\frac{d + \frac{\alpha}{\nu}}{d}\right) \tag{10}$$

where $A$ represents the laser absorptivity coefficient, $P$ defines the laser power [W], $\kappa$ represents the thermal conductivity at [W/m.K], $T_b$ represents the boiling temperature [K], $d$ expresses the laser beam size [m], and $v$ describes the scanning speed [m/s].

### Processing informed lack-of-fusion criterion

Various criteria proposed in the literature for lack-of-fusion, balling, and keyholing evaluate regions in the *process spaces* where the defects may occur.

**Table 7 | We applied these equations and labels for each respective defect to graph the printability maps in Fig. 5**

| Defect Type | Label | Equation |
|---|---|---|
| Lack-of-Fusion | LOF1 | $D \leq t$ |
| | LOF2 | $(\frac{h}{W})^2 + \frac{th}{th+D} \geq 1$ |
| Balling | Ball1 | $\frac{L}{W} \geq 2.3$ |
| | Ball2 | $\frac{\pi W}{L} < \sqrt{\frac{2}{3}}$ |
| Keyholing | KH1 | $\frac{W}{D} \leq 2.5$ |
| | KH2 | $\frac{\Delta H}{h_s} = \frac{AP}{\pi h_s \sqrt{\alpha v a^3}} > \frac{\pi T_{boiling}}{T_{liquidus}}$ |
| | KH3 | $Ke = \frac{\eta P}{(T_{liquidus} - T_0)\pi\rho C_p \sqrt{\alpha v r_0^3}} > 6$ |

In addition to criteria that describe defects in processing parameter spaces, indicators that are solely functions of *alloy chemistry* also exist.

For lack-of-fusion, we leverage two sets of criteria (Equations (11) and (12)), where $t$ is the powder layer thickness. Lack-of-fusion pores occur when insufficient energy is deposited onto the material, which results in incomplete bonding with the underlying layers. Therefore, if the melt pool depth is less than the powder layer thickness, the melt pool will not fully bond to the substrate and previous layer, leading to a lack of fusion pores. Equation (11) captures this geometric constraint[14]. Equation (12) defines the second criterion for lack-of-fusion introduced by Zhu et al.[7]. According to this criterion, lack-of-fusion may occur when the hatch spacing, $h$, exceeds the maximum hatch spacing, $h_{max}$[7,15], beyond which porosity due to insufficient overlapping melt tracks occurs. The hatch spacing must provide a good join between adjacent tracks to avoid incomplete bonding and to prevent lack-of-fusion-induced porosity. Equation (13)[15] calculates the maximum hatch spacing and allows us to estimate the maximum hatch spacing that would still result in fully dense layers at different locations of the process map.

$$D \leq t \tag{11}$$

$$\left(\frac{h}{W}\right)^2 + \frac{th}{th + D} \geq 1 \tag{12}$$

$$h_{max} = W\sqrt{1 - \frac{t}{t + D}} \tag{13}$$

### Processing-informed balling criteria

Balling occurs when the melt pool breaks into droplets instead of forming a continuous pool due to Plateau-Rayleigh capillary instability. The instability arises at high laser power and scan speeds and can be evaluated using a criterion based on the ratio of the melt pool length and width ($L/W$), as defined in Equation (14). The second criterion evaluates balling using the approach proposed by Yadroisev et al.[94]. For single tracks, their analysis defines the threshold value between a melt pool's stability zone and instability zones using Equation (15).

$$\frac{L}{W} \geq 2.3 \tag{14}$$

$$\frac{\pi W}{L} < \sqrt{\frac{2}{3}} \tag{15}$$

### Processing informed keyhole criteria

At combinations of high power and low velocity, rapid evaporation of the molten liquid forms keyhole-induced pores by causing a deep penetration of the molten material through recoil pressure on the melt pool surface, which creates vapor cavities. Consequently, the resultant melt pool exceeds the depth that develops during the conduction mode. Equation (16)[6,15] suggests an experimentally-obtained criterion in terms of melt pool geometry.

However, King et al.[12] introduced another criterion for keyholing that considers not only the geometry of the melt pool but also the material properties and processing parameters. They showed a positive correlation between the normalized enthalpy and the melt pool depth, which correlates to keyholing. Equation (17) defines the criterion using the specific enthalpy, $h_s = \rho C_p T_{liquidus}$ ($\rho$ the density [kg/m$^3$], $C_p$ defines the effective specific heat [J/kg K], and $T_{liquidus}$ denotes the liquidus temperature). In the original formulation, King et al.[12] considered the heat capacity for the solid phase. However, the heat capacity for the solid phase underestimates the energy absorbed by the metal from the laser. Therefore, we replaced the heat capacity with the effective heat capacity to account for both the sensible and latent heat of the given alloy from room temperature until the liquidus temperature.

Recently, Gan et al.[21] introduced a 'universal' keyholing criterion that defines regions of conduction, transition, and keyholing mode in the melt pool. This dimensionless keyholing criterion, $Ke$, as defined in Equation (18), derives from dimensional analysis and a modification of the Buckingham-Pi theorem. The criterion $Ke$ utilizes processing parameters such as the laser power ($P$ [W]), velocity ($v$ [m/s]), and beam radius ($r_0$ [m]) as well as material properties, i.e., the absorptivity ($\eta$), liquidus temperature ($T_{liquidus}$ [K]), substrate temperature ($T_0$ [K]), density ($\rho$ [kg/m$^3$]), heat capacity ($C_p$ [J/kgK]), and thermal diffusivity ($\alpha$ [m$^2$/s]). In their work, Gan et al.[21] used experiments to derive a threshold value of $Ke > 6.0$ for the keyhole mode in the L-PBF processes.

$$\frac{W}{D} \leq 2.5 \tag{16}$$

$$\frac{\Delta H}{h_s} = \frac{AP}{\pi h_s \sqrt{\alpha v a^3}} > \frac{\pi T_{boiling}}{T_{liquidus}} \tag{17}$$

$$Ke = \frac{\eta P}{(T_{liquidus} - T_0)\pi\rho C_p \sqrt{\alpha v r_0^3}} > 6 \tag{18}$$

### Summary of processing-informed criteria

Our framework considers two criteria for lack-of-fusion, two for balling, and three for keyholing, resulting in a total of twelve different combinations of these defect criteria. Table 7 summarizes the criteria for lack-of-fusion, balling, and keyholing. When constructing a printability map, we also plot contour lines, which define the maximum hatch spacing allowed to prevent porosity due to a lack of overlap of the melt pool tracks (defined in Eq. (13)) on the printability maps.

The shifting boundaries observed across different printability maps in Supplementary Material for a single composition and thermal model arise from the specific combinations of defect criteria. For example, replacing KH1 with KH2 introduces a new keyholing boundary, which affects the melt pool depth, and modifies boundaries for defects like lack-of-fusion (e.g., LOF1) as the depth is updated based on the G-S model. These shifts highlight the sensitivity of the framework to criteria selection, as each criterion incorporates unique assumptions or parameters. This sensitivity to defect criteria furthermore motivates ensembling all 12 defect criteria into a single probabilistic defect map, illustrated in Fig. 6.

### Composition-informed hot cracking susceptibility

Hot cracking, also known as hot tearing, forms fissures in casting due to solidification shrinkage and thermal strains arising during solidification and subsequent cooling[95]. Sufficient liquid feeding to the mushy zone and high grain growth mitigate hot cracking. This defect typically occurs during the late stages of solidification, when restricted liquid feeding and slower grain

growth increase susceptibility[13,95]. Researchers have identified several indicators of hot cracking, which they have applied in alloy design[96,97]. One common hot cracking indicator, the solidification range, measures the difference between an alloy's solidus and liquidus temperatures. Studies have shown that wider solidification ranges increase the tendency for hot cracking[98] and compositional microsegregation during alloy solidification[99].

Furthermore, several hot cracking criteria build on the Scheil-Gulliver curve, which describes how the liquid phase fraction varies with temperature ($f_L - T$) during non-ideal solidification. For example, Clyne and Davies[95] developed a hot cracking susceptibility coefficient, which is the ratio between the time a casting remains vulnerable to cracking, $t_v$, and the time it can relieve thermal stresses, $t_R$[95,100], as shown in Equation (19). Hot cracking typically occurs during the late stages of solidification, when restricted liquid feeding heightens susceptibility. Estimating these critical timescales involves applying a Scheil-solidification simulation to generate heat-evolution vs. temperature and liquid fraction vs. temperature curves. Assuming $dQ/dt \propto 1/\sqrt{t}$ results in a liquid fraction vs. time curve ($f_L - t$), which determines the critical timescales $t_v$ and $t_R$. Clyne and Davies[100] established a convention in which $t_R$ represents the period during which $f_L$ is between 0.6 and 0.1, whereas $t_v$ represents the period during which $f_L$ is between 0.1 and 0.01[100]. Larger $t_v/t_R$ ratios indicate an increased frequency of hot cracking[95,100].

$$CSC = \frac{t_v}{t_R} \tag{19}$$

### Chemistry-informed balling susceptibility

Balling occurs when an elongated melt pool is unstable and segregates into disconnected beads or balls[101,102]. Processing parameters and material properties influence balling. By focusing on melt pool geometry[94] or the intrinsic material properties of the printed material[101], simplified models provide a way to describe this phenomenon.

Regarding material properties, Zhou et al.[101] used a simple model to predict balling in tungsten, where melt spreading and solidification compete. When the characteristic spreading time ($\tau_{spread}$) exceeds the characteristic solidification time ($\tau_{solid}$), solidification likely arrests melt spreading before the melt pool successfully spreads and wets the underlying surface, resulting in balling[101]. Both of these timescales are functions of intrinsic material properties. In a previous work[73], we further developed this concept by using $\tau_{solid}/\tau_{spread}$ as a physics-based indicator to predict the presence of balling in an L-PBF specimen. An AM database consisting of 2527 unique alloy-process observations and 158 unique compositions validated the predictive ability of the indicator. Analysis revealed that a larger $\tau_{solid}/\tau_{spread}$ correlates with a smaller balling region in the printability map associated with an alloy.

This model relies on several assumptions. The model treats the spreading and solidification of the droplet separately. When conduction dominates the heat transfer mode, the characteristic solidification time consists of two distinct regimes, $\tau_1$ and $\tau_2$. In the first regime (Eq. (20)), $\tau_1$ represents the time conduction takes to remove heat from a spherical droplet with radius $a$ via conduction to the underlying substrate. The second regime (Eq. (21)) describes the time to remove the latent heat of fusion $L$ and solidify. Equation (22) provides the total solidification time, where the factor of 2 accounts for the fact that approximately half of the droplet's surface contacts the underlying substrate, halving the heat transfer area and approximately doubling the solidification time. Assuming isothermal and homologous wetting of a molten droplet on a flat surface, with solidification and oxidation neglected, capillary flow primarily drives spreading. Equation (23) estimates the characteristic spreading time[101]. L-PBF processes typically assume the initial radius of the droplet, $a$, to be 100 $\mu$m[73,101]. In addition, assumptions treat the fusion temperature, $T_f$, as distinct and defined as the material's solidus temperature[73], and the ambient temperature, $T_a$, as 25 °C.

$$\tau_1 = \frac{a^2 k}{3\alpha k_a} \ln\left(\frac{T_o - T_a}{T_f - T_a}\right) \tag{20}$$

$$\tau_2 = \frac{a^2 k}{3\alpha k_a}\left(1 + \frac{k_a}{2k}\right)\frac{L}{C(T_f - T_a)} \tag{21}$$

$$\tau_{solid} = 2(\tau_1 + \tau_2) \tag{22}$$

$$\tau_{spread} = \left(\frac{\rho a^3}{\sigma}\right)^{1/2} \tag{23}$$

### Construction and ranking of the printability maps

After obtaining the melt pool profiles from one of the three thermal models previously mentioned, the criteria listed in Table 7 determine regions in the process space associated with lack-of-fusion, balling, and keyholing. Different combinations of criteria define these defect regions. These defect regions construct printability maps. The criteria sets that construct these maps benchmark their performance by comparing the predictions in the printability maps with the available data.

To rank the printability maps and evaluate the various criteria combinations, we assume the combination of models and criteria functions as a binary classifier separating a specific defect region from the rest of the processing space—e.g., lack-of-fusion or not lack-of-fusion. The experimental data points corresponding to a particular label represent the positive class, while everything else represents the negative class. To rank the performance, a wide variety of performance metrics, such as precision (Eq. (24)), recall (Eq. (25)), and accuracy (Eq. (26)), evaluate points in the positive (P) and negative (N) classes. Furthermore, TP stands for true positive or the number of positive classes that were correctly classified; TN stands for true negative or the number of negative points that were correctly classified; FP stands for false positive or the number of negative points that were misclassified, and FN stands for false negative or the number of positive points that were misclassified.

Each printability map uses the metrics defined in Eqs. (24)–(26) for each of the four binary decision cases. The metrics are then averaged to find the average performance of each set of criteria. The averaging considers the prevalence of each type of defect within the experimental data set. From now on, we refer to the weighted performance metric as (i.e., average accuracy will be referred to as accuracy), assuming that each metric discussed has already been averaged to fully assess the performance of the printability maps. Accordingly, we place experimental points on each constructed printability map to facilitate the visualization and calculation of the performance metrics mentioned above. The Supplementary Material includes the precision and recall values for each map.

$$Precision = \frac{TP}{TP + FP} \tag{24}$$

$$Recall = \frac{TP}{TP + FN} \tag{25}$$

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{26}$$

### Ensemble probability printability map

The 12 printability maps, created using the criteria listed in **Summary of Processing-Informed Criteria**, provide insight into the material's printability by highlighting areas of consensus and divergence between the criteria for the respective defect in the processing space. These maps serve as valuable tools for discerning the intricacies of printability, where areas of agreement indicate a higher probability of having a correct prediction of the defect

occurring, while discrepancies hint at underlying uncertainties. Consolidating the information from all 12 printability maps creates a single ensembled probabilistic printability map. However, emphasizing the more accurate printability maps in the final ensemble is crucial. The softmax function in Equation (27) determines and assigns the weights for each map. In Equation (27), $x$ represents the weighted average accuracy for each of the 12 printability maps, and $w$ represents the weight factor. We employ the softmax function to normalize the accuracies for each of the 12 printability maps and convert the accuracies into model probabilities that sum to 1. This approach ensures that each map contributes proportionally to the ensemble printability map.

The weight factor helps determine the likelihood of each criteria set, and the experimenter adjusts the weight factor based on their analysis and a priori knowledge of the criteria set. However, caution is necessary when assigning a high weight factor, $w$, to favor one criteria set over others. Consequently, substantial empirical data remains essential for establishing boundaries for defect regions and effectively biasing one criteria set over others. Forecasting the model probability linked with each criteria set aids in understanding the uncertainty inherent in the criteria set and printability map, helping to identify the optimal criteria set for a given alloy.

Utilizing the 12 printability maps crafted through the criteria outlined in **Summary of Processing-Informed Criteria**, we delve deeper into material printability analysis by identifying consensus and divergence points within the processing space's defect criteria. These maps serve as invaluable tools in unraveling printability intricacies, where areas of alignment indicate higher confidence in defect predictions while disparities underscore underlying uncertainties. Consolidating insights from all 12 maps, we construct an ensembled probabilistic printability map. To achieve this, we employ the softmax function (Eq. (27)) to ascertain weights for each map. Here, $x$ represents the weighted average accuracy across the 12 printability maps, and $w$ denotes the weight factor. By utilizing the softmax function, we normalize map accuracies and convert them into model probabilities, ensuring a proportional contribution from each map to the ensemble.

$$\sigma(x)_i = \frac{e^{w*x_i}}{\sum_{i=1}^{n} e^{w*x_i}} \tag{27}$$

### Inclusion & Ethics statement

Our study on a computational framework to predict printability maps for L-PBF manufactured alloys adheres to ethical standards, ensuring transparency in research methodologies and data handling. We are committed to promoting diversity and inclusivity in computational materials science, fostering an environment that values equity and accessibility in our research endeavors.

## Data availability

The primary data supporting this study's conclusions are provided within the manuscript and with the GitHub repository. Additional datasets can be accessed upon request from the corresponding author.

## Code availability

The underlying code for this study is publicly available at https://github.com/sofiasheikh05/Printability_Map_Framework/.

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## Author contributions

S.S. led the creation of the manuscript, with critical contributions to various sections by B.V. and P.H. A.M.K. developed the NN surrogate code for the E-T Model. P.M. and D.S. provided support in compiling the code. I.K., A.E., and R.A. assisted in conceptualizing and guiding both the manuscript and the concepts presented. All authors participated in editing and reviewing the manuscript.

## Competing interests

The authors declare no competing interests.

## Additional information

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